CHEST RADIOGRAPH

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1) DIFFERENT VIEWS.

2)BASIC CHEST X RAY INTERPRETATION.

3)VARIOUS BASIC PATHOLOGIES.

DIFFERENT VIEWS

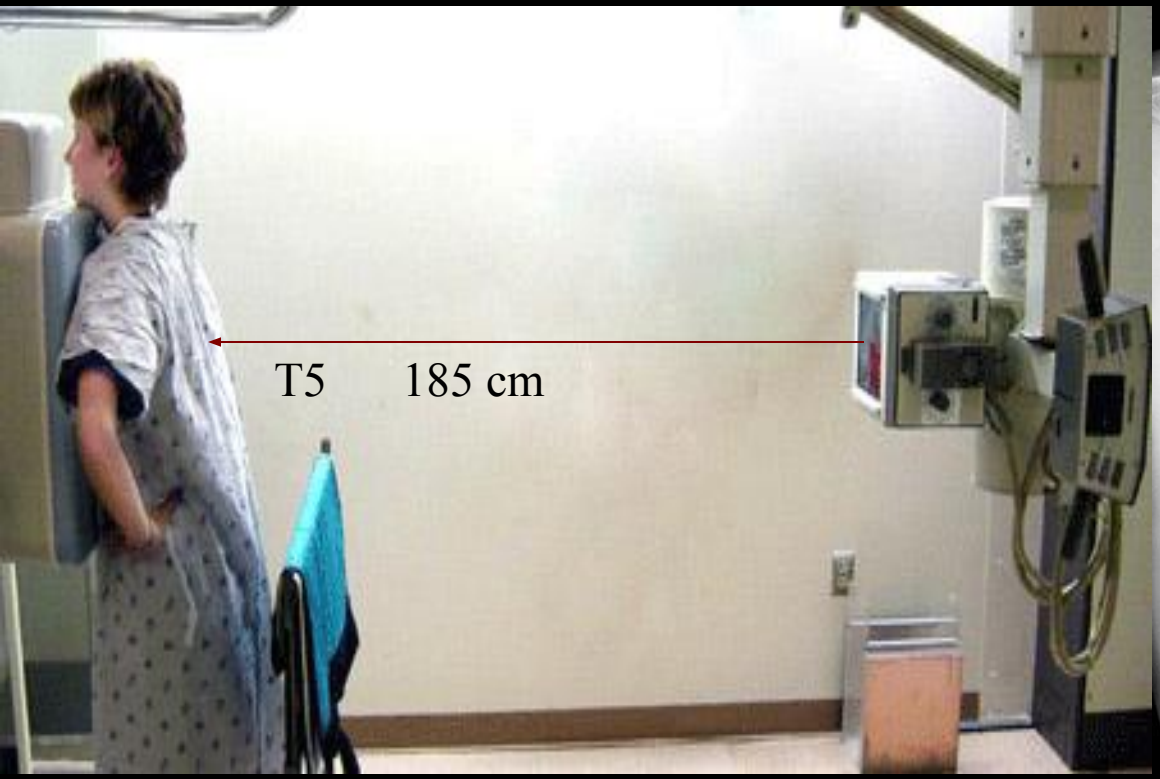
BASIC - PA VIEW-ERECT.

ALTERNATE - AP VIEW -ERECT.
-SUPINE.
-SEMIERECT

SUPPLYMENTARY- LATERAL VIEW [erect & decubitus].
- LORDOTIC.
- PA VEIW EXPIRTION

Surface dose for standard patient is 0.3mGy

PA VIEW





- Patient position.
- Centering of x ray beam.
- FFD (film to focus distance)
- Low kVp tech(60-70 kVp) –Adequate for PA view, lungs, ribs & if not using grids.
- High kVp tech(120-150 kVp) –medistinum and heart.
- Exposure time < 40ms and factors mAs – 16-20

PA view

- Standard view for evaluation of chest.
- Patient stands with chin up and shoulders rotated forward facing the film.

- Exposure is made on full inspiration centring at T_5 .
- Low kvp –calcification,pleural plaque ,pulmonary nodule are visualised excellent in low kvp film.
- High kvp used for increased visualisation of hidden areas, airways, vascular structure etc

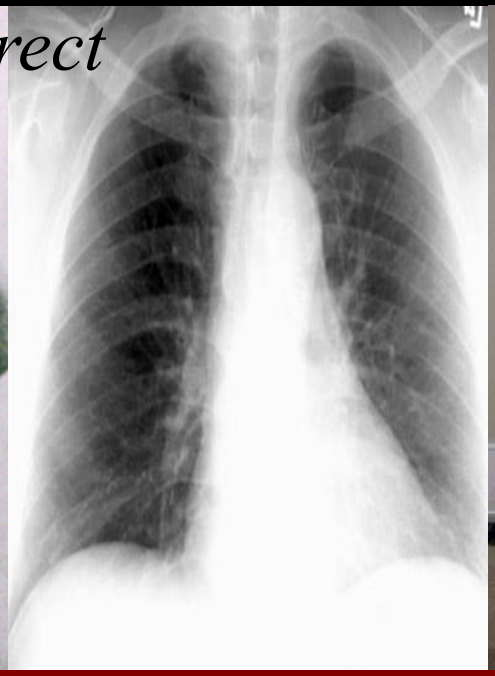
AP VIEW

- Sometimes the only projection possible in sick patients, unable to stand
- Helpful in deciding whether the small pulmonary opacity on PA view is genuine by altering its relationship to overlying ribs and vascular shadows.

AP VIEW.

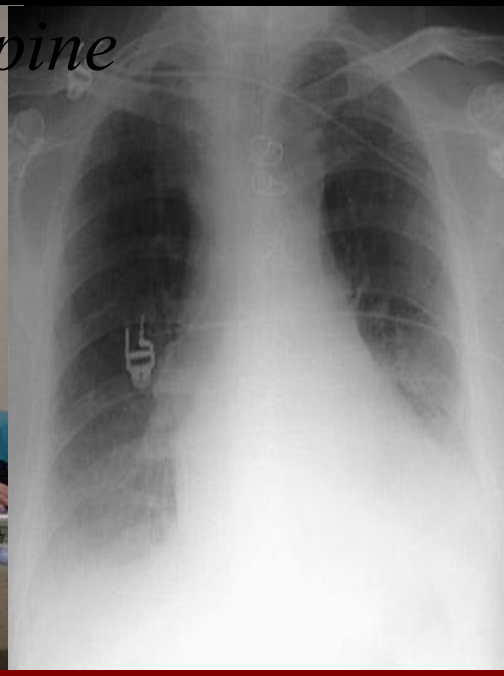
Erect





Supine

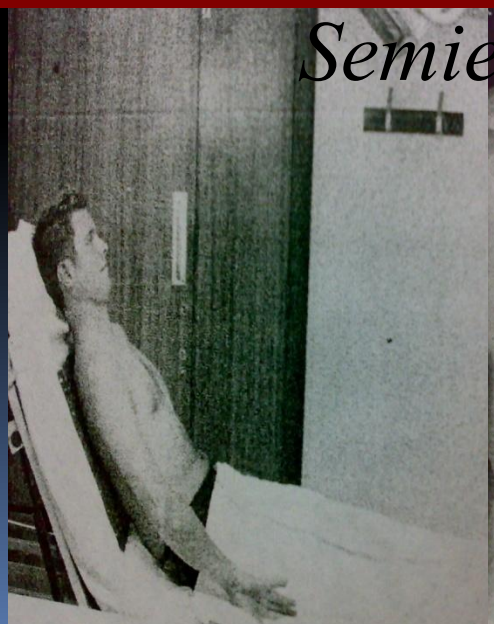




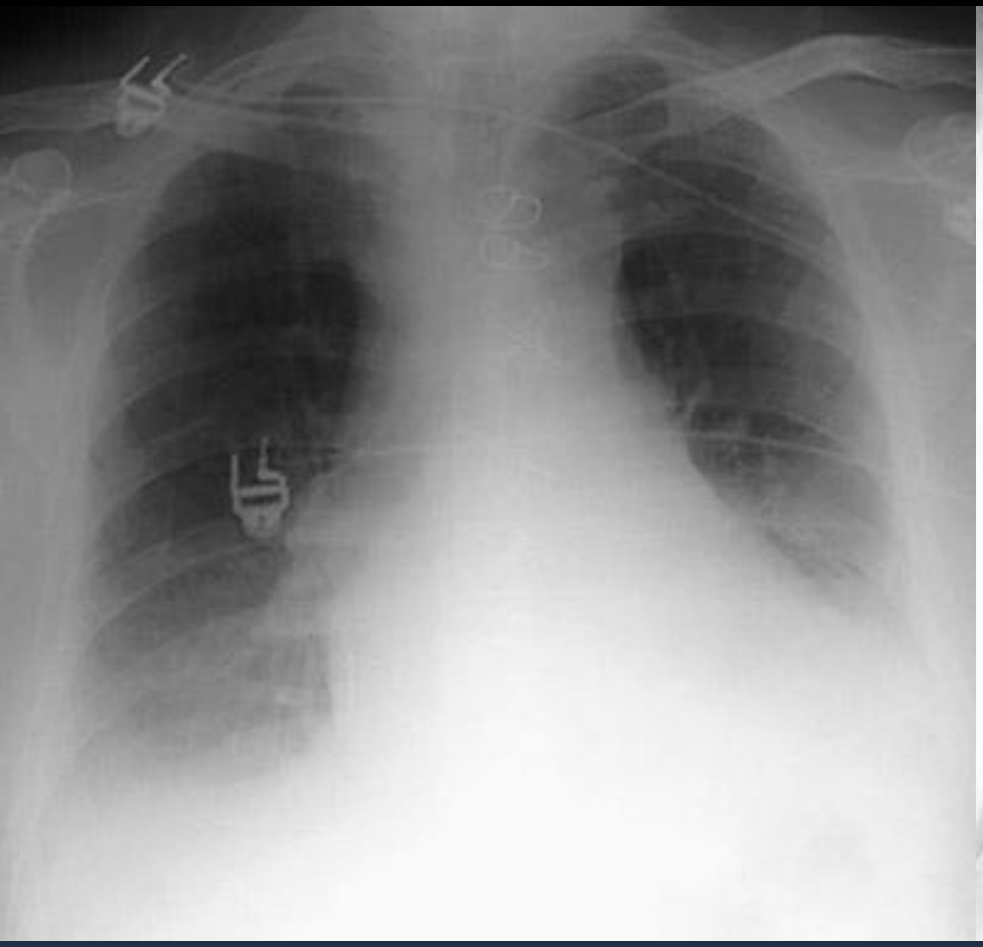
- Patient position.
- Centering of x ray beam.

- FFD (film to focus distance) – 100 cm
- mAs 80-100

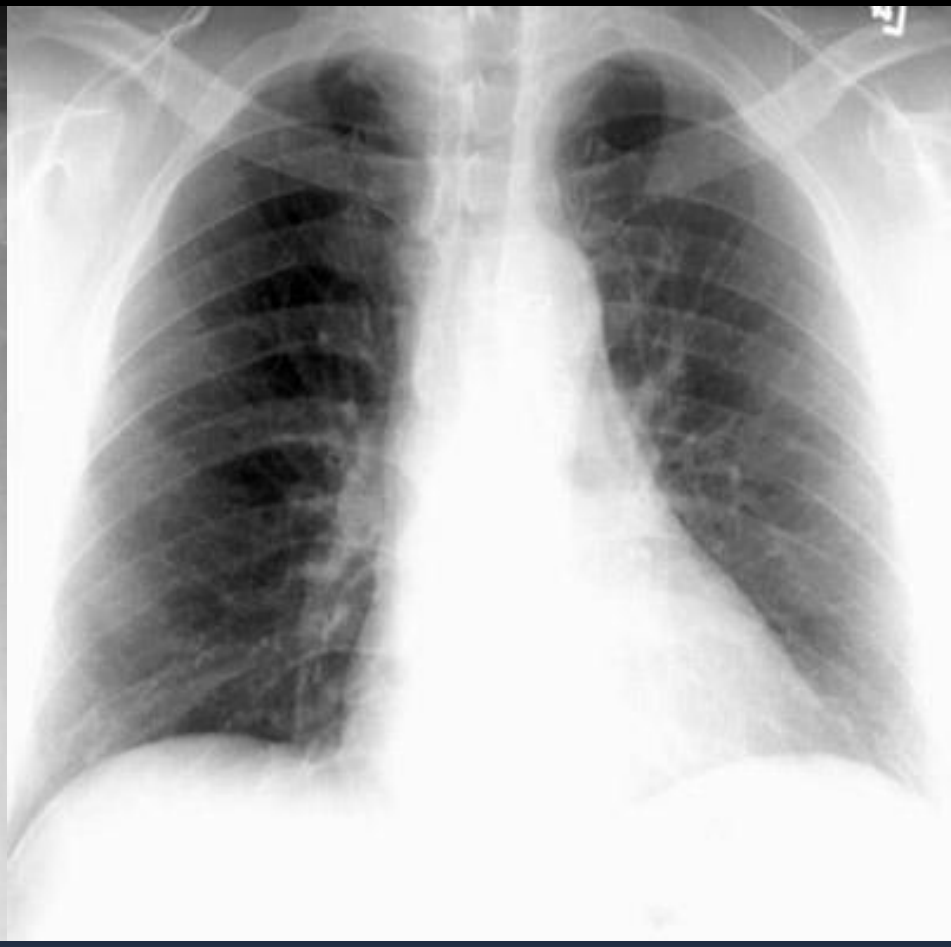
Semierect







AP

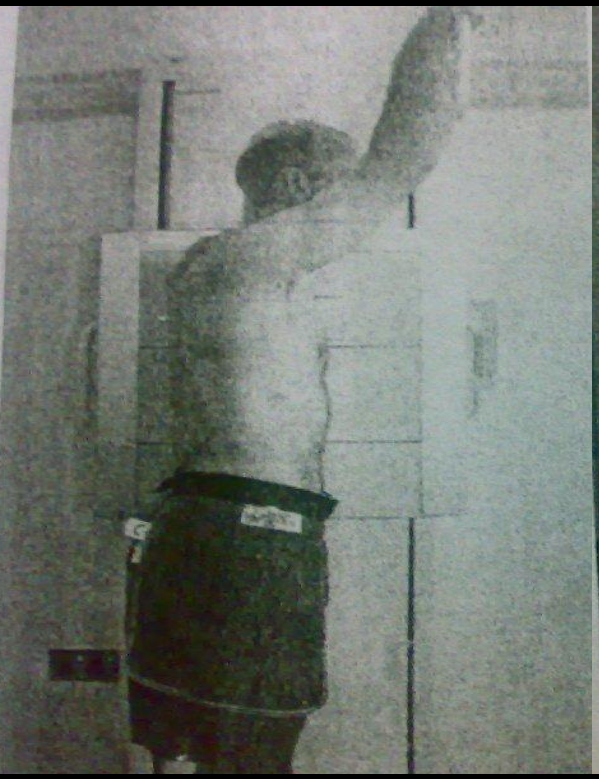


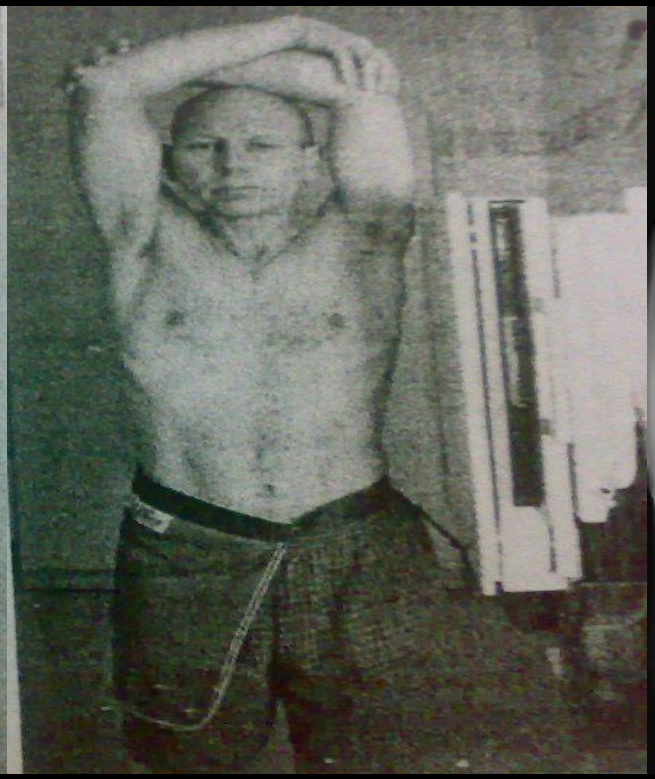
PA

DIFFERENCE BETWEEN AP & PA VIEW

AP VIEW	PA VIEW
Clavicles are projected more cranialy over the apices	Projected more oblique
Cardiac shadow more magnified	Less magnified
Scapulae overlie lung fields	Scapulae do not overlie lung fields
Ribs are more horizontal	less horizontal
Vertebral spines are less prominently seen	More prominently seen

LATERAL – ERECT.







- Patient position.
- Centering of x ray beam.
- FFD (film to focus distance) 6 ft
- mAs 32-48.

LATERAL VIEW -ERECT

- Lesions obscured on PA view are clearly demonstrated on lateral view (ant mediastinal masses ,post basal consolidation,encysted pleural fluid etc)
- Helpful in localisation of different lobes and segments,
- Patient stands with side of interest nearer the film, arms are elevated, forearms are resting on the head.

LATERAL - DECUBITUS.





- Patient position.
- Centering of x ray beam.
- kV 50-60 , mAs 20 .

LATERAL DECUBITUS

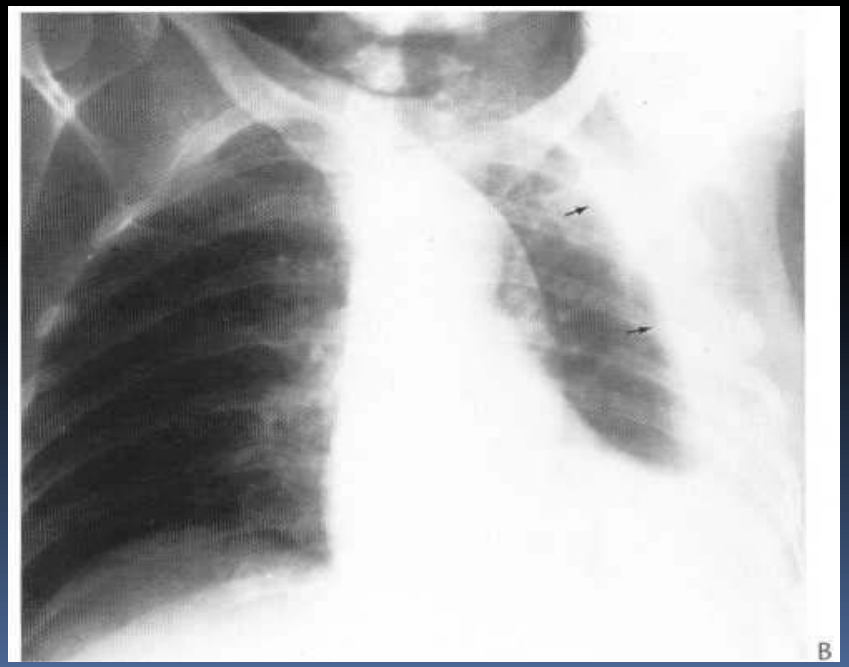
- Shows the fluid level particularly well.
- Helps to diagnose small pnumothorax or pleural effusion
- Patient lies in lateral recumbant position (lie in affected side), arms extended above the head, film is placed in standing position in front of the chest.
- Patient lies in same position for five minutes to allow the fluid trickle down in dependent part of chest.

DECUBITUS VIEW

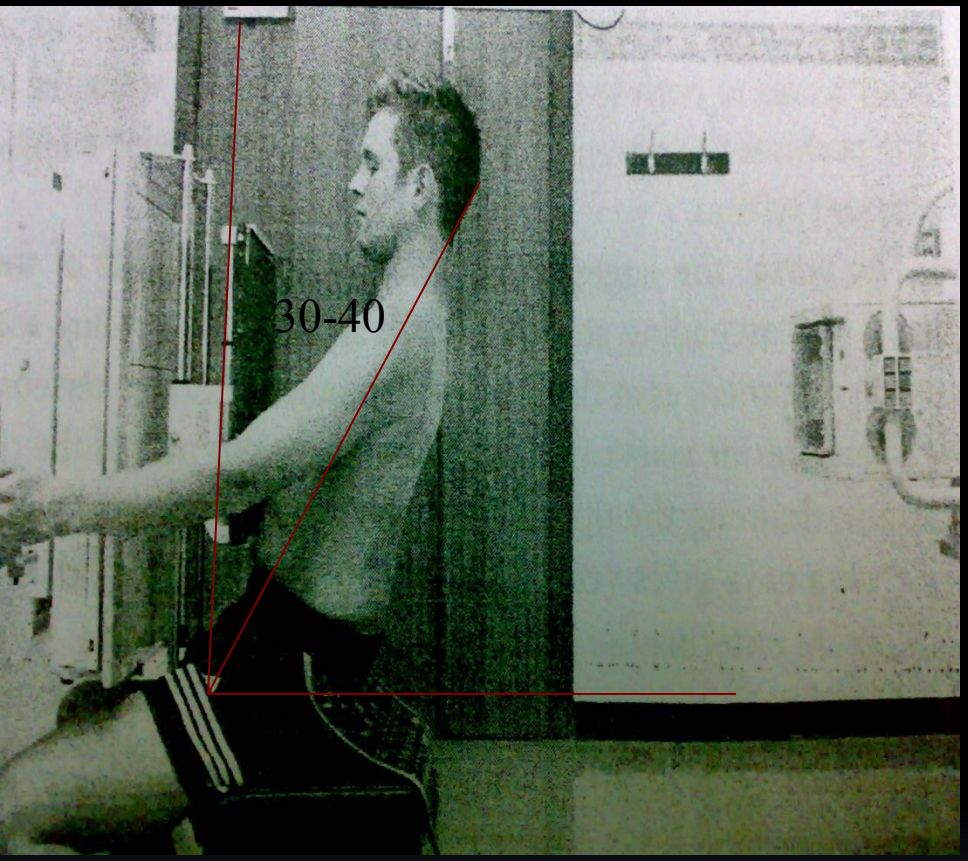
- Mild elevation of the diaphragm on PA view

- Fluid along the left lateral chest wall on decubitus view





LORDOTIC.





- Patient position.
- Centering of x ray beam.
- Used to demonstrate Posterior Apical Disease, middle lobe collapse, mediastinal herniation and inter lobar pleural effusion.

LORDOTIC view

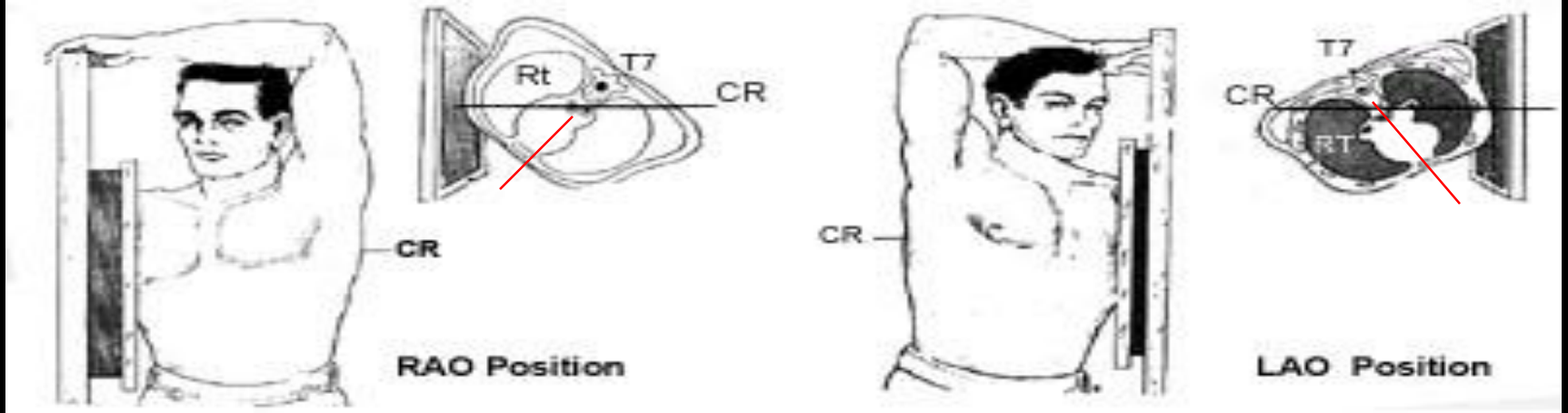
- Used to demonstrate Posterior Apical Disease, middle lobe collapse, mediastinal herniation and inter lobar pleural effusion.
- Patient leans back in lordotic position or with cranial tilt of tube
- Clavicles projected above the lung field.

OBLIQUE VIEWS

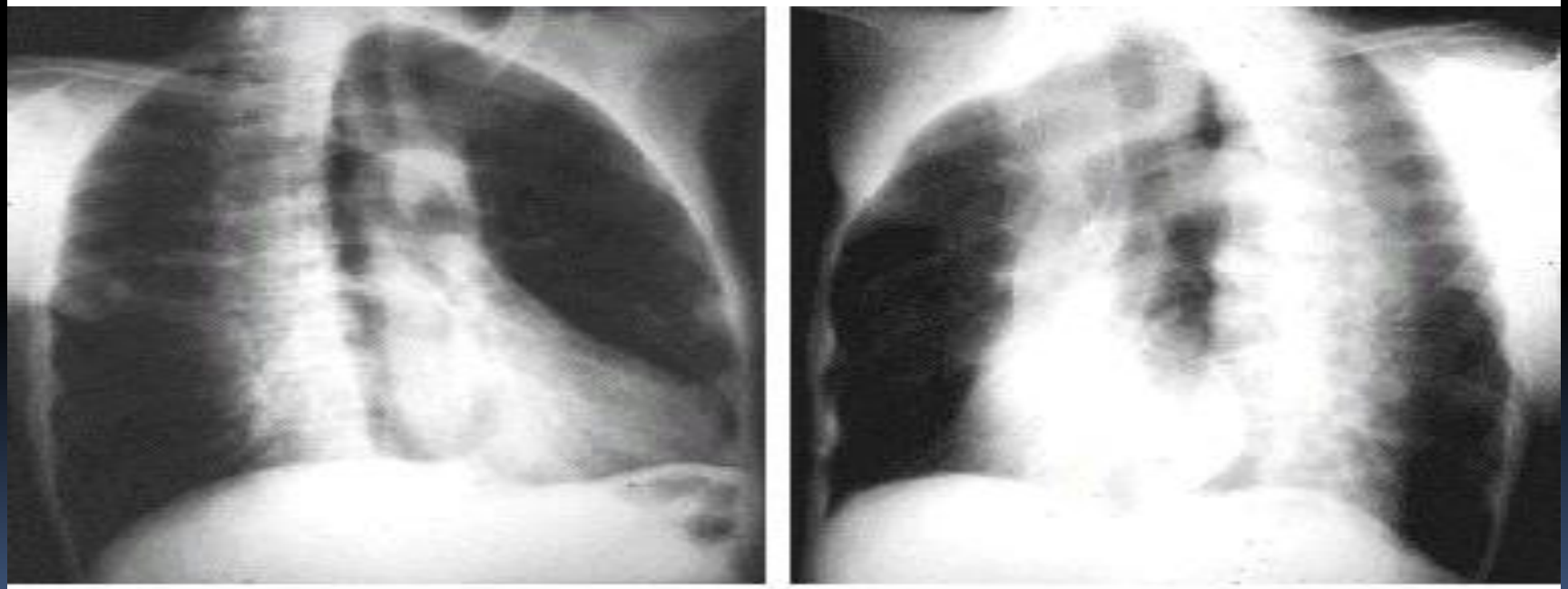
Retrocardiac area
Posterior CP angles
B/L Lung disease – No overlap

RAO - Rt. Anterior Oblique
LAO - Lt. Anterior Oblique

RPO - Rt. Posterior Oblique
LPO – Lt. Posterior Oblique



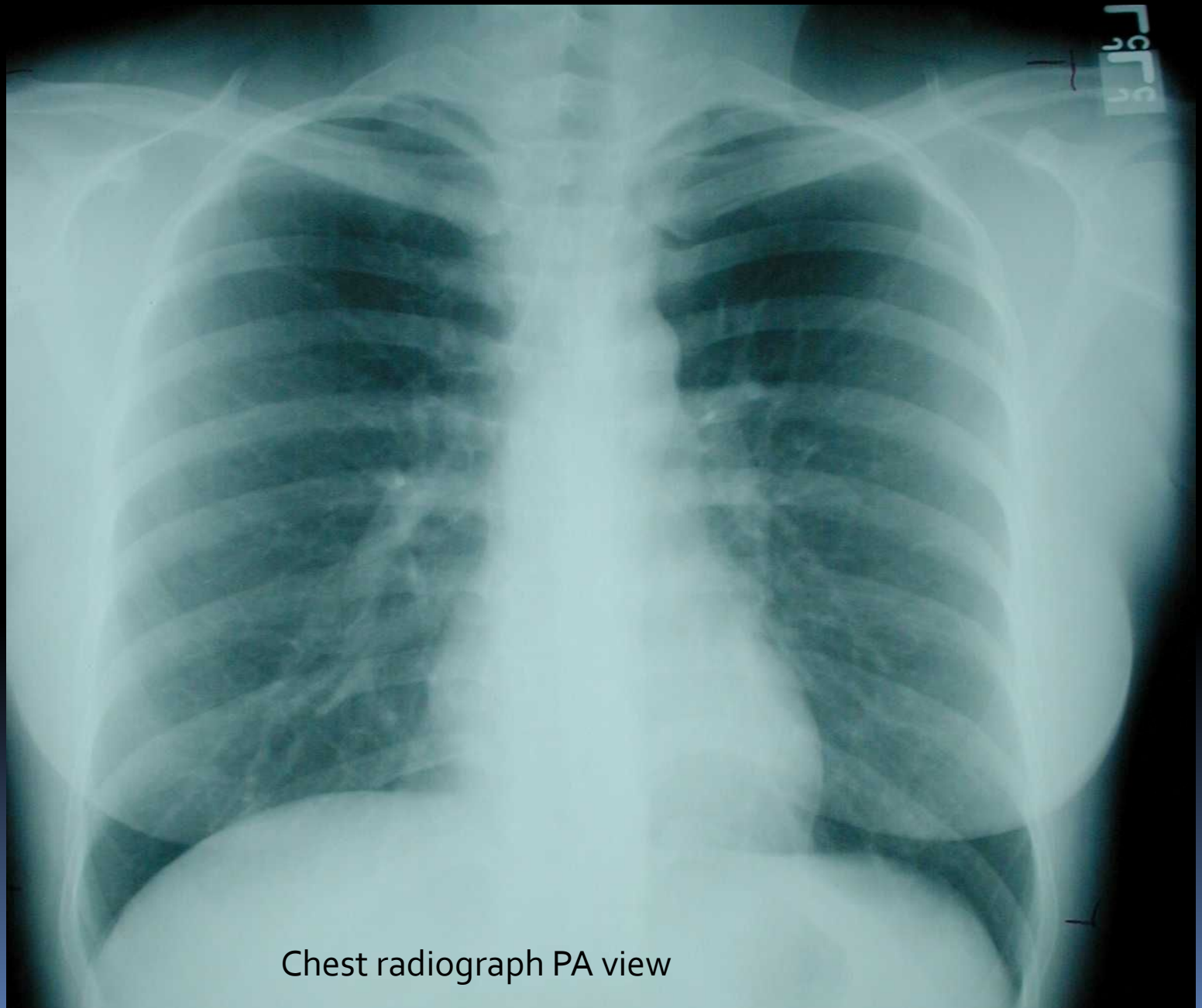
Anterior (PA) oblique projections -- the side of interest is the side furthest from the cassette. Image courtesy of Dr. Naveed Ahmad.



A right anterior oblique (right image) and left anterior oblique (left image) view demonstrate maximum area of the lung field. Image courtesy of Dr. Naveed Ahmad.

Basic Chest X-Ray Interpretation

Have a system



Chest radiograph PA view

Basic Chest X-Ray Interpretation

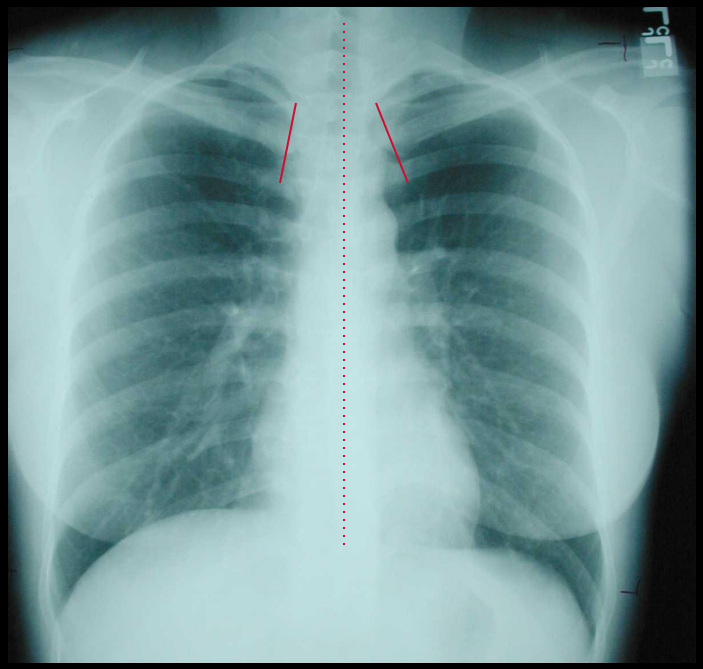
- Identification
- Side marker
- Positioning
- Penetration
- Inspiratory film
- Visualized structures

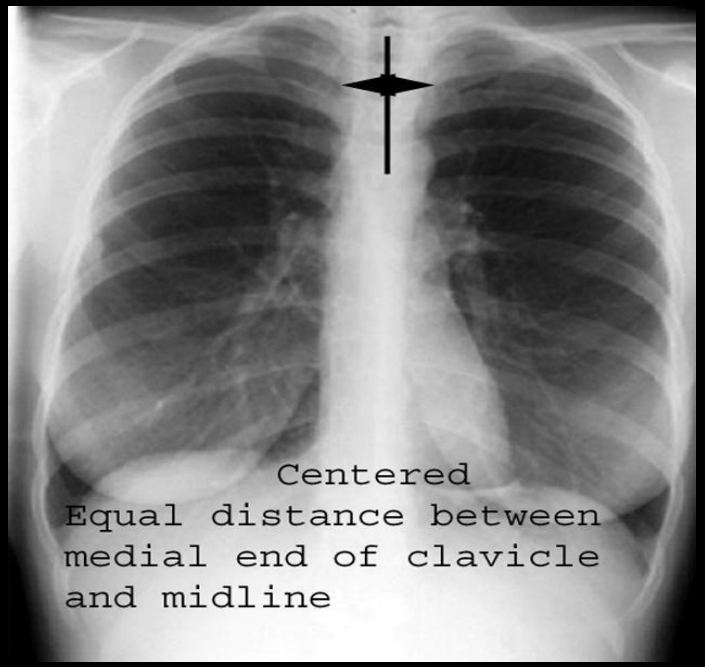
SIDE MARKER

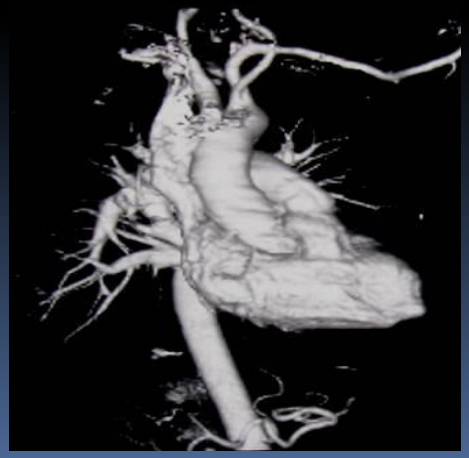
Orientation of the aortic arch, gastric bubble and heart should be determined to confirm normal situs and that the side markers are correct.

POSITIONING

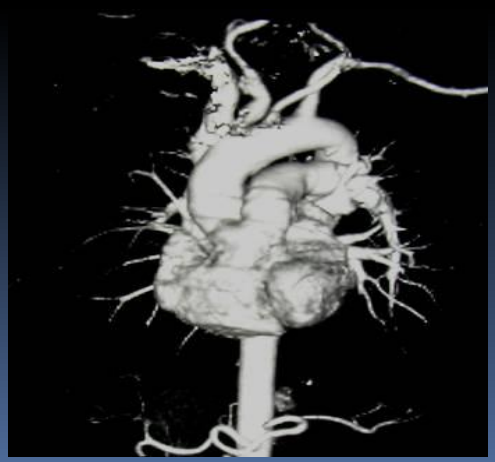
straight vs oblique











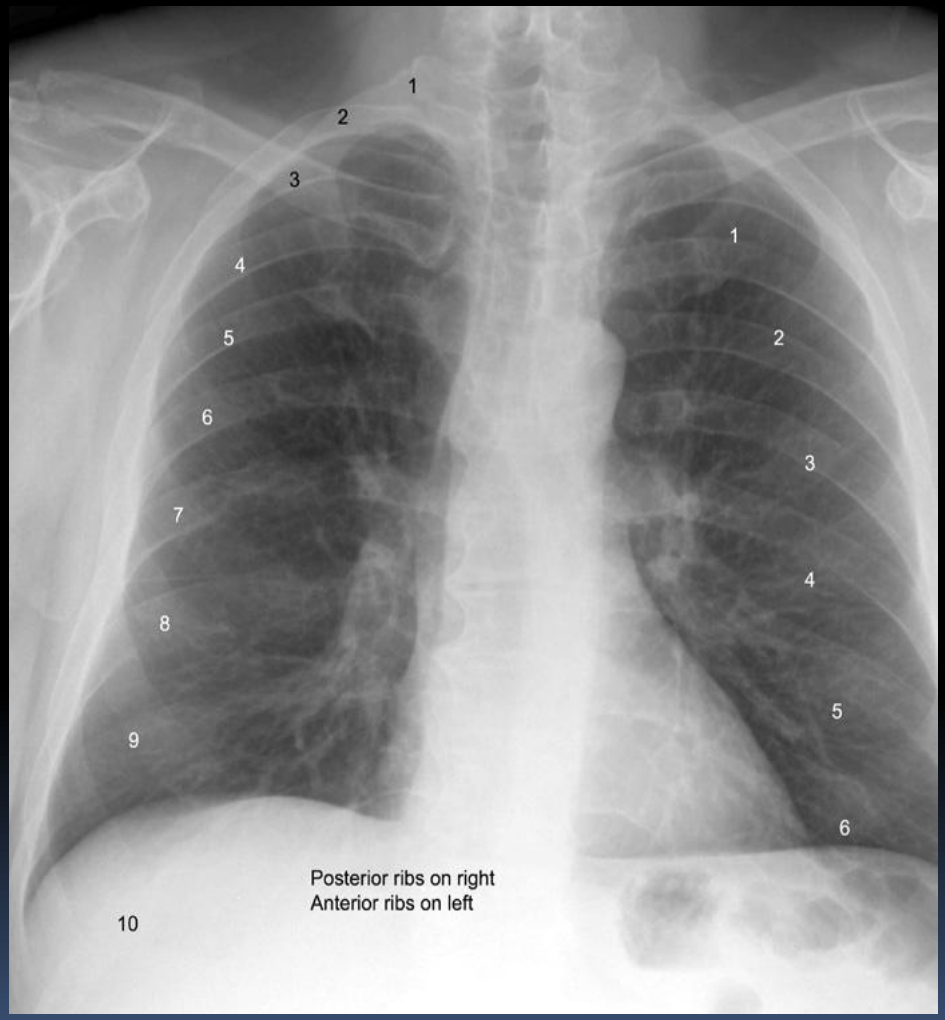
PENETRATION

vertebral bodies and disc spaces should be just visible down to t8/t9 level through cardiac shadow

underpenetration

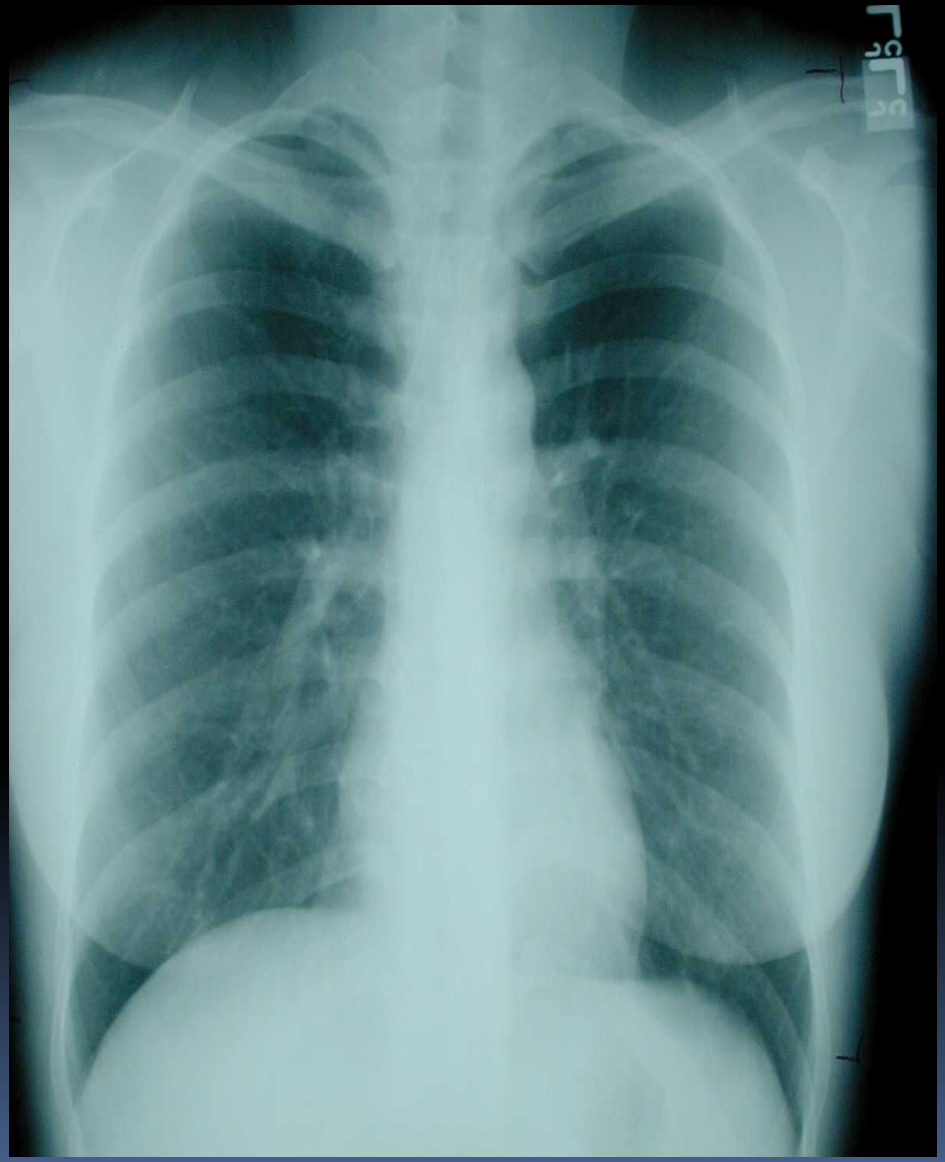
overpenetration

COUNTING OF RIBS



posterior anterior

VISUALISED STRUCTURES

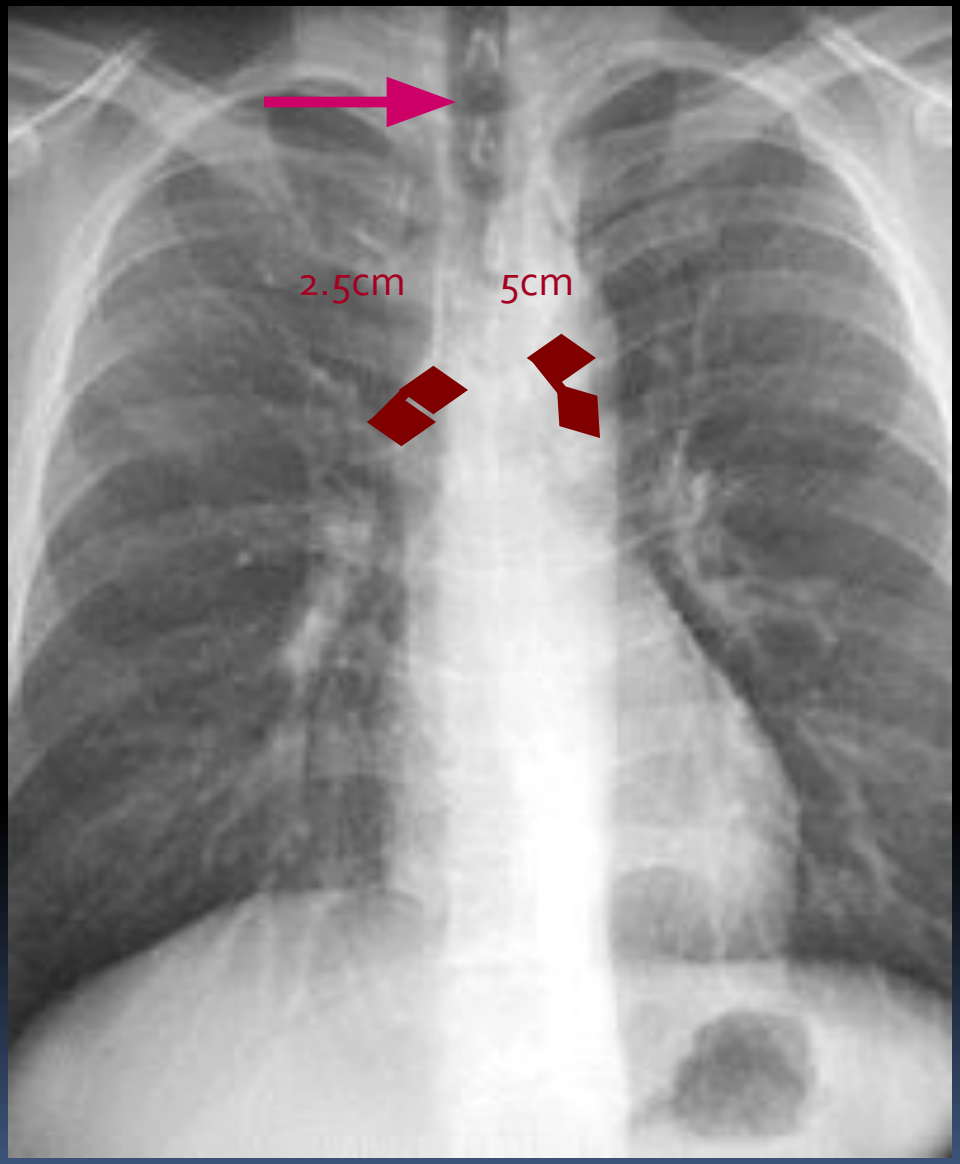


TRACHEA

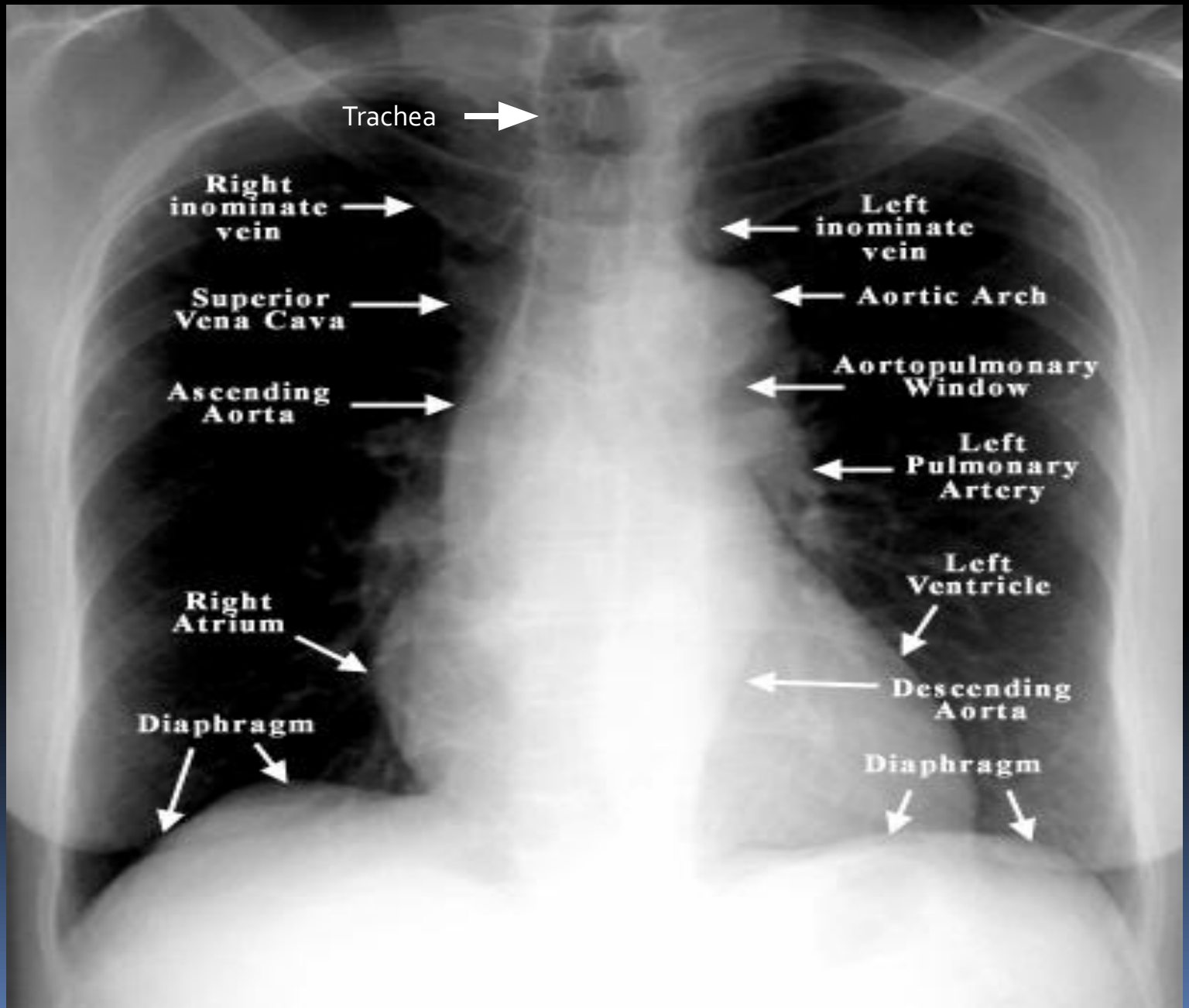
- Trachea should be examined for narrowing, displacement and intraluminal lesions.
- In midline ,deviate slightly to right at around aortic knuckle
- Divides into rt and lt bronchus at T5 .rt bronchus is short and straight,
- Normal carinal angle is 60-75 degree
- Right paratracheal stripe< 5mm.

Trachea

- Position
- Outline



Systematic approach

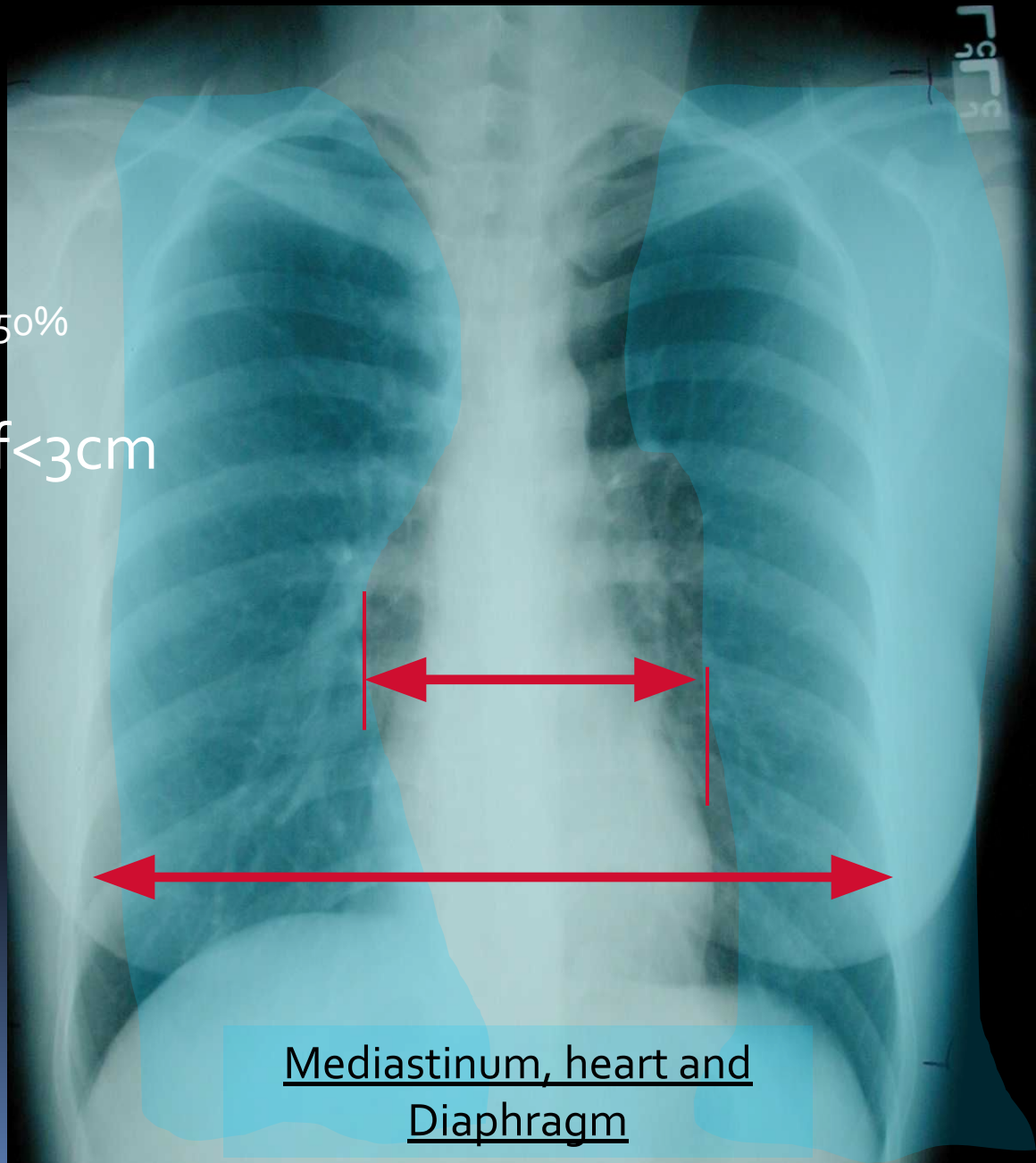


HEART, MEDIASTINUM AND DIAPHRAGM

Cardiac shadow-2/3rd left

Cardiothoracic ratio <50%

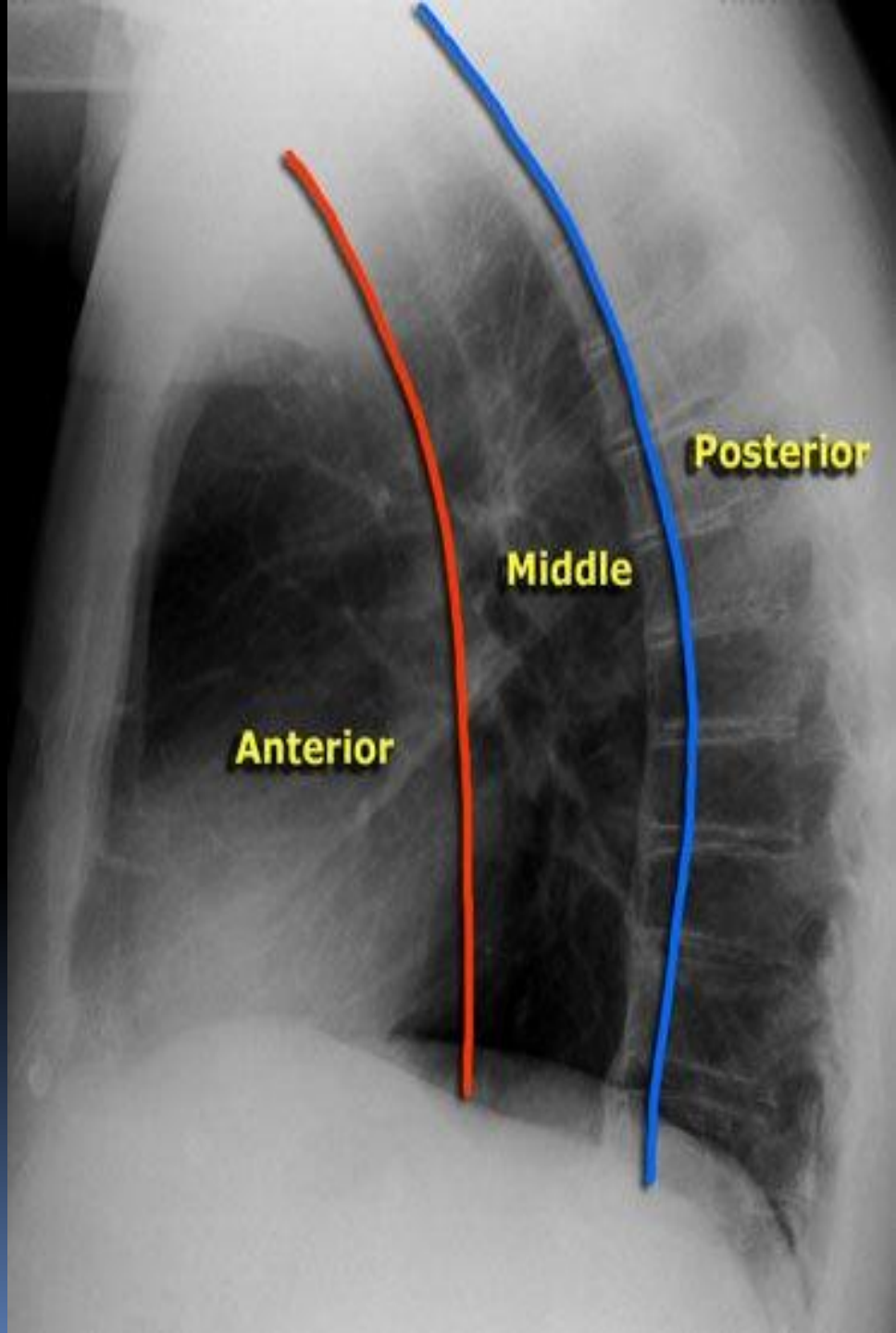
Diaphragm-rt higher,diff<3cm



Mediastinum, heart and Diaphragm

MEDIASTINUM

1. Anterior
2. Middle
3. Posterior



FISSURES

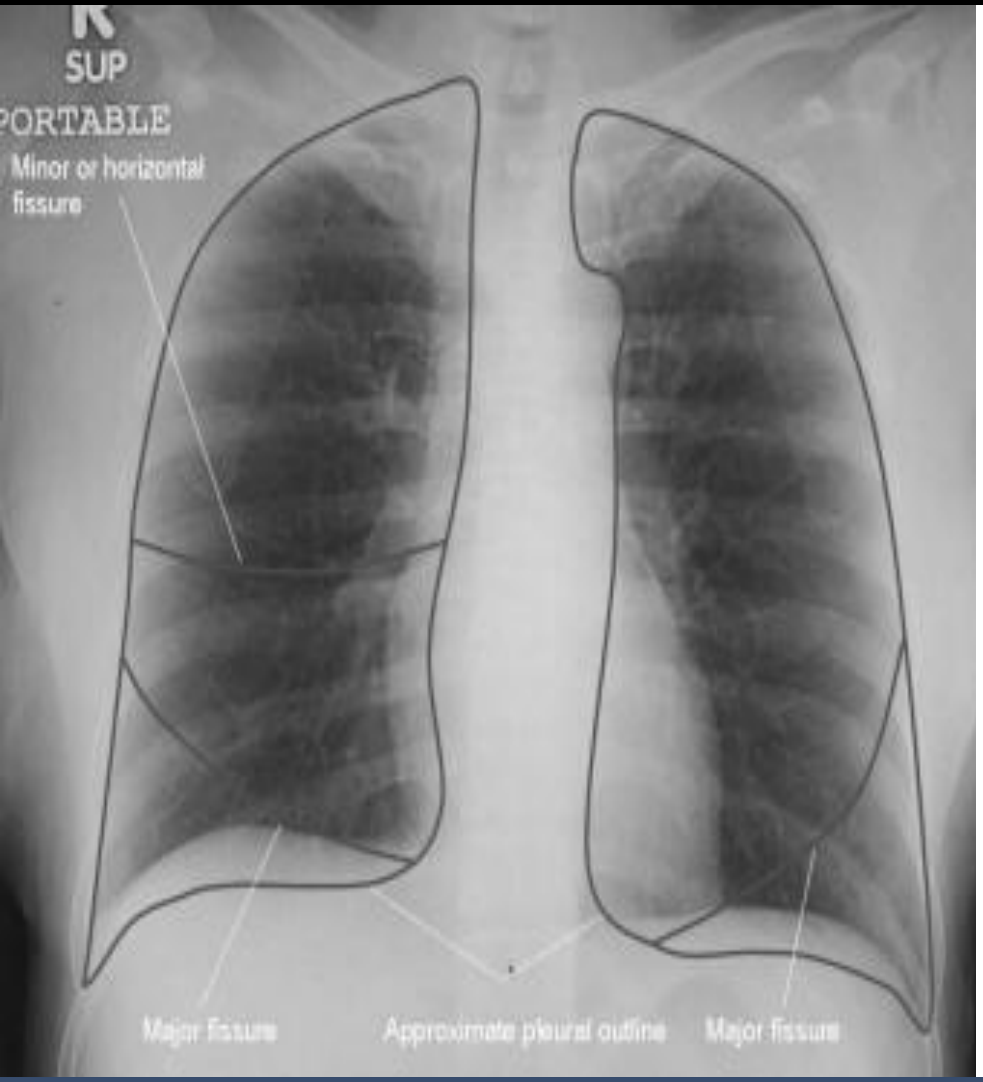
- PA-horizontal run from hilum to sixth rib in axillary line
- Lat- horizontal fissure-run anteriorly
- Both oblique commence posteriorly at T4 ,pass through hilum ,left finishes 5 cm behind and rt jus behind ant cp angle.

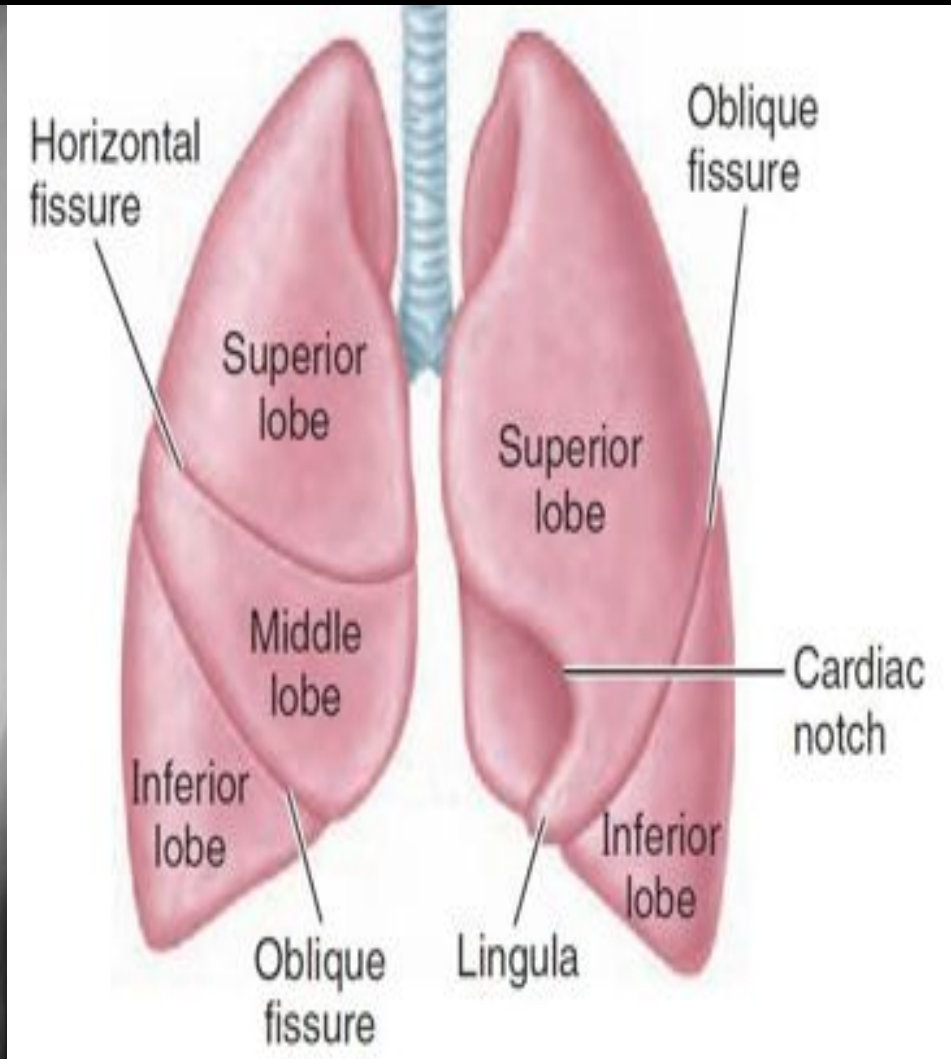
FISSURES

- Main fissures
  1. Oblique
  2. Horizontal

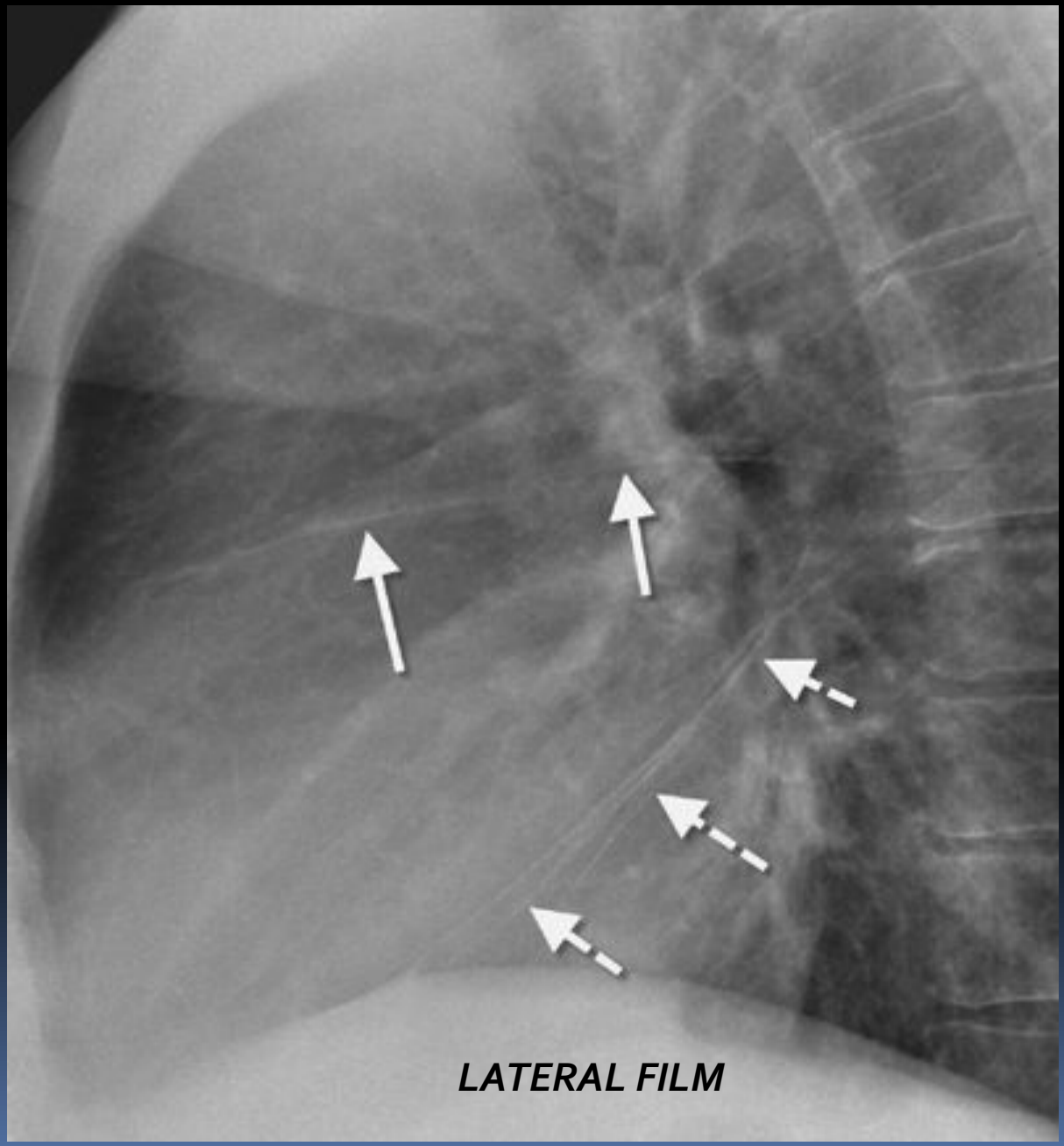
- Accessory fissures
  1. Azygous fissure (mc)
  2. Superior accessory fissure
  3. Inferior accessory fissure
  4. Left side horizontal fissure

Fissures


SUP
PORTABLE
Minor or horizontal fissure
Major Fissure
Approximate pleural outline
Major fissure


Horizontal fissure
Oblique fissure
Superior lobe
Superior lobe
Middle lobe
Cardiac notch
Inferior lobe
Inferior lobe
Oblique fissure
Lingula

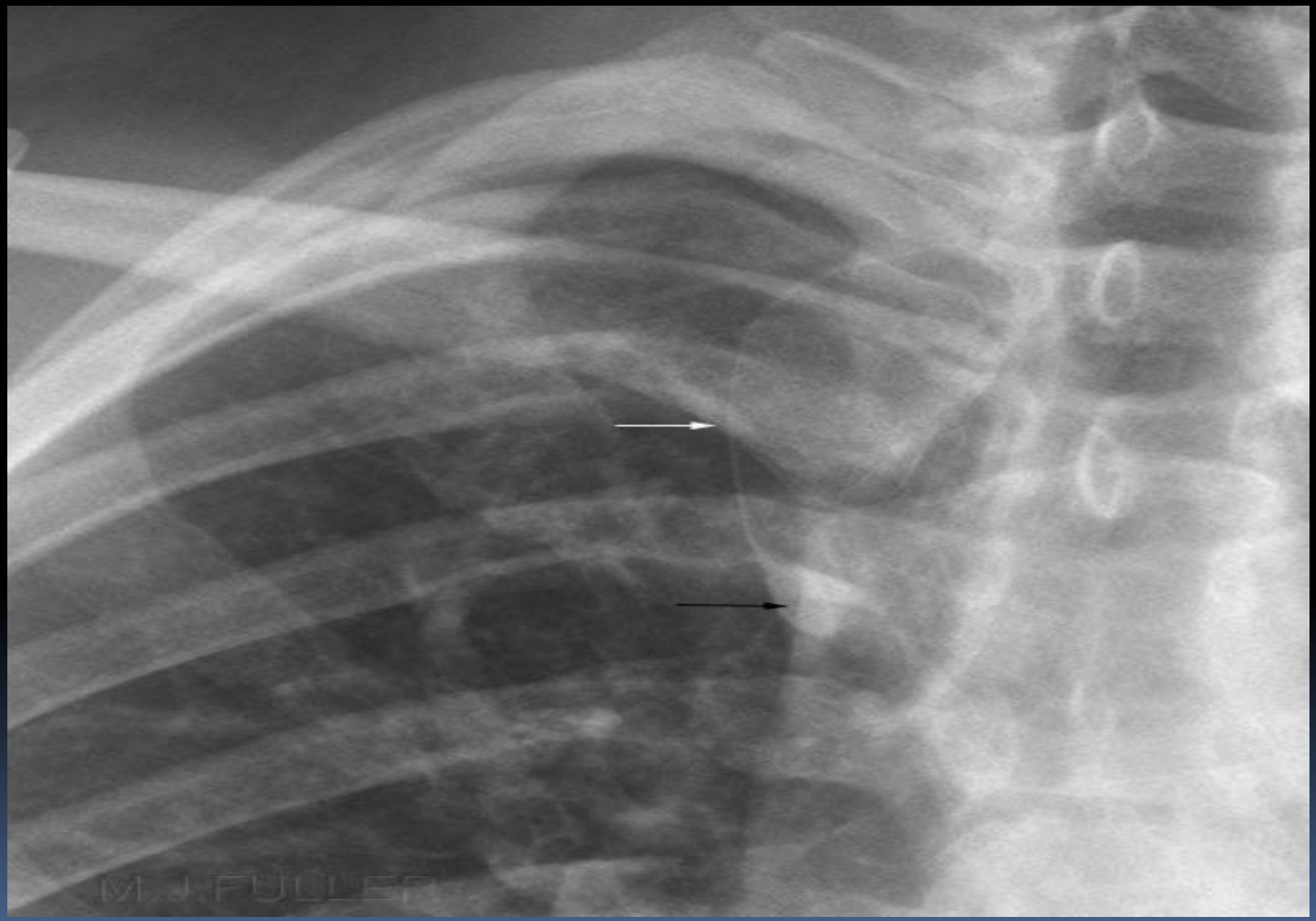
FISSURES

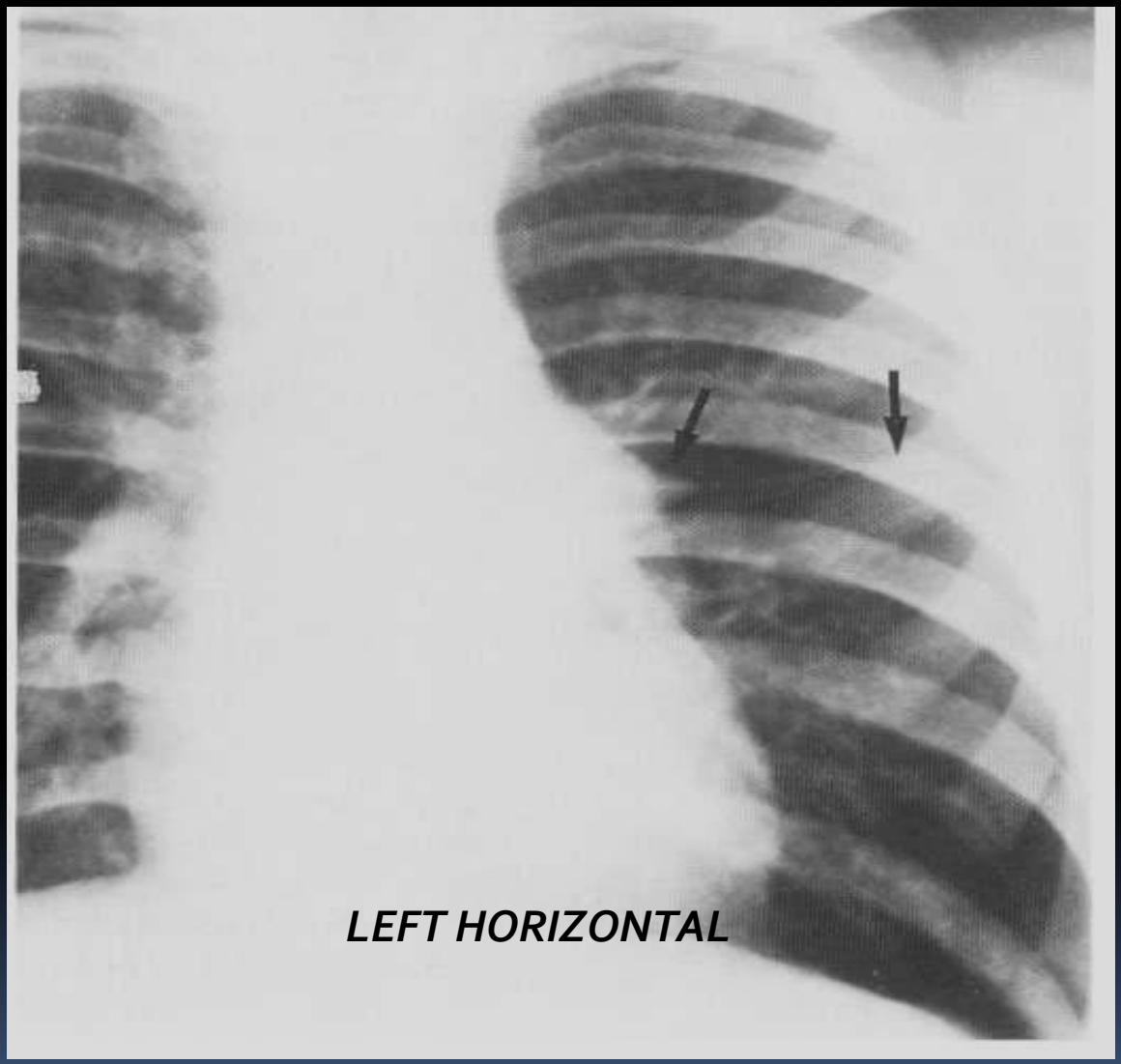
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ACCESSORY FISSURES

- AZYGOUS FISSURE- comma shaped, right sided,It forms in apex of lung and consist of paired folds of visceral and parietal pleura plus the azygous vein which has failed to migrate.
- SUPERIOR ACCESSORY FISSURE - separate apical from basal segment of lower lobe, resembles horizontal fissure on PA view ,on LAT runs posteriorly from hilum.
- INF ACCESSORY FISSURE -oblique line running cranially from CP angle to hilum, separate medial basal from other basal segments.
- LEFT HORIZONTAL FISSURE -separate lingula from upper lobe segments

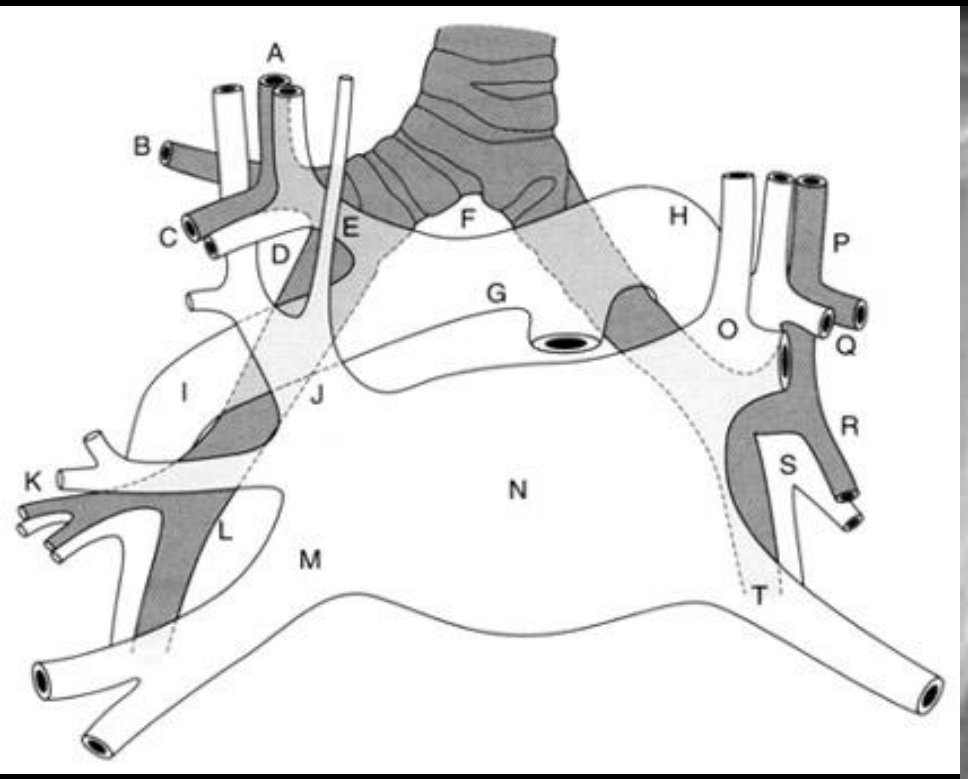
AZYGOUS FISSURE





LEFT HORIZONTAL

Hilum

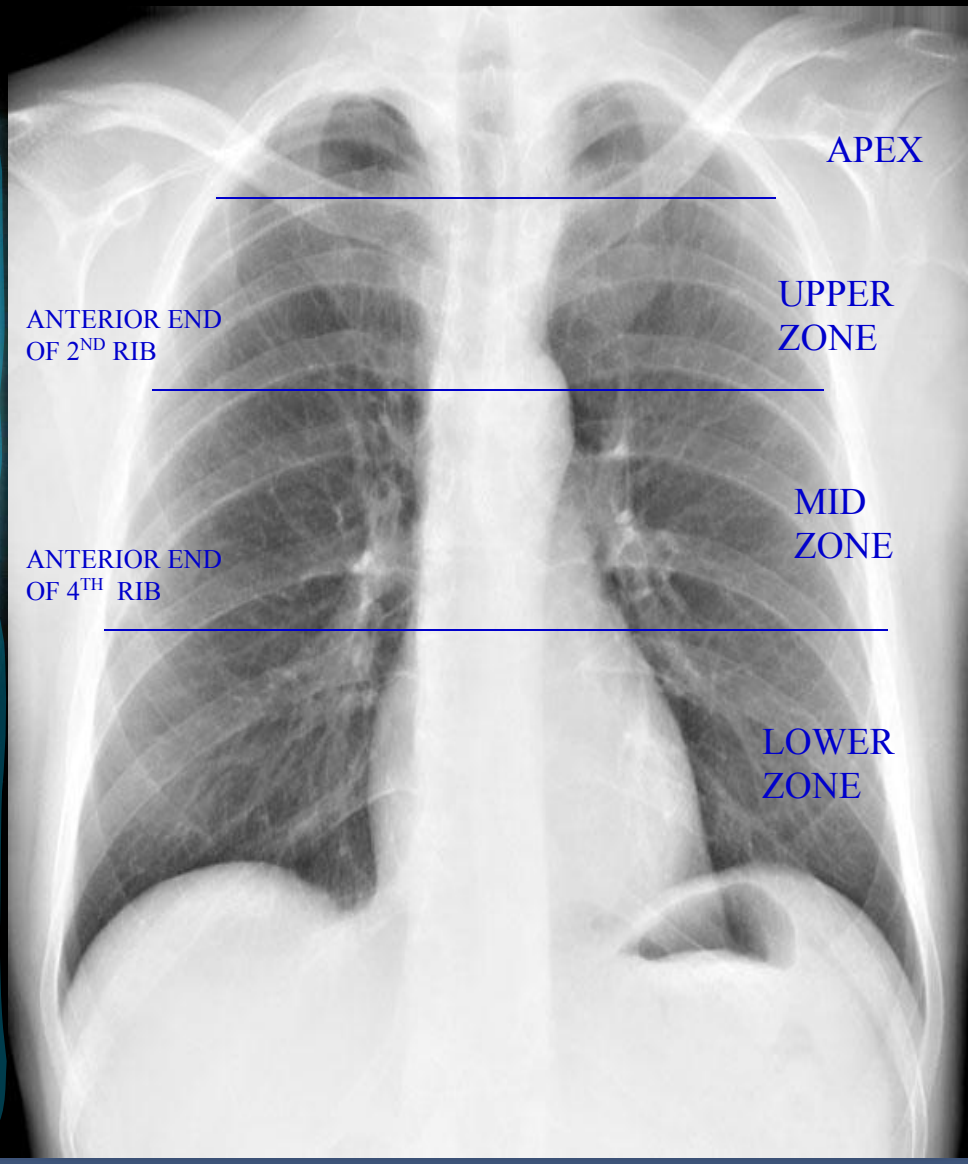




- 97% left higher than right
- Pulmonary vessels &upper lobe veins
- Equal size and density
- Bronchial vessels
- Lymphatic system

LUNGS



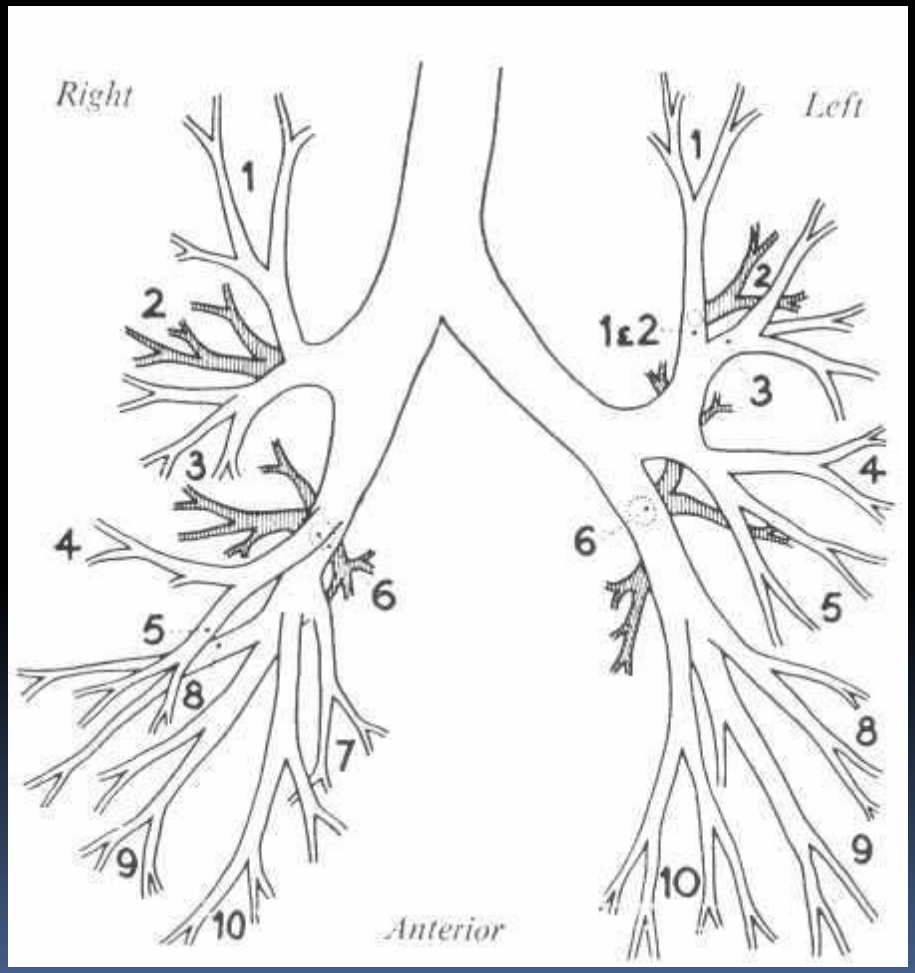


RADIOGRAPHIC ZONES ON CHEST RADIOGRAPH

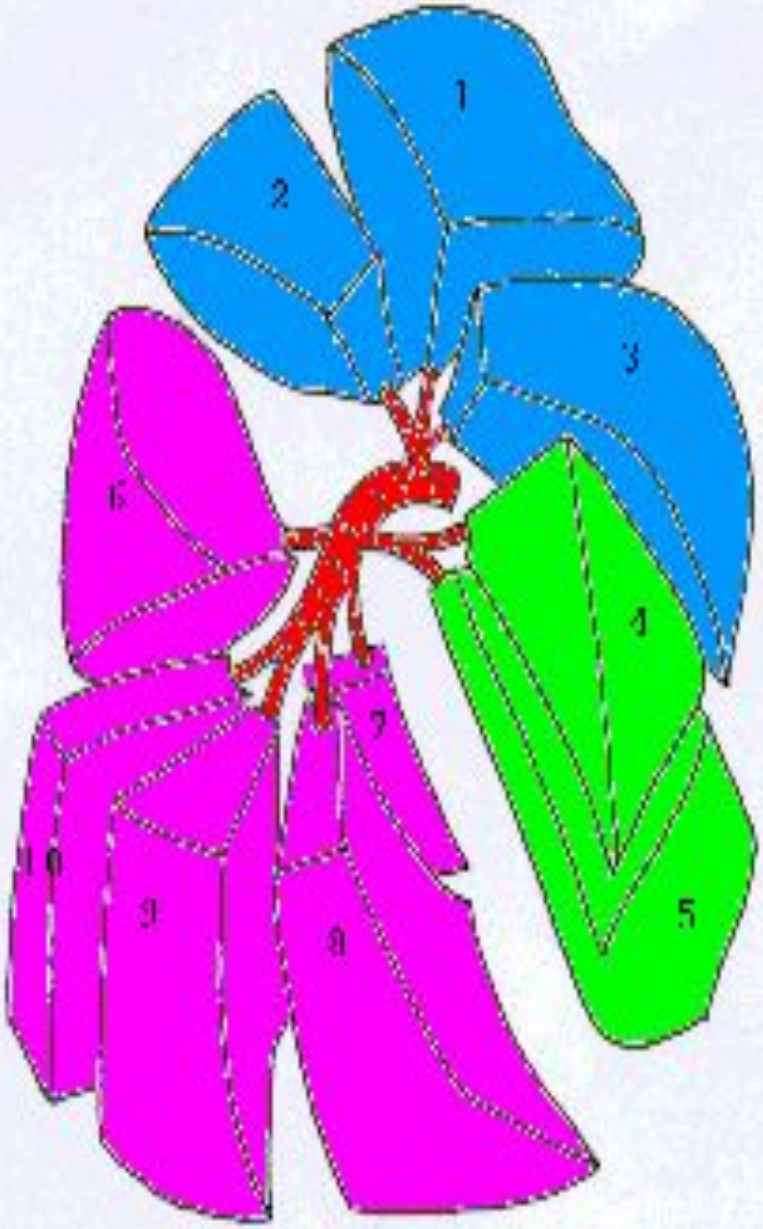
SEGMENTS

- Lungs are divided into three lobes on right and two lobes on left side,
- Bronchopulmonary segments are based on the subdivisions of the lobar bronchi.
- Supplied by segmental bronchi and vessels.

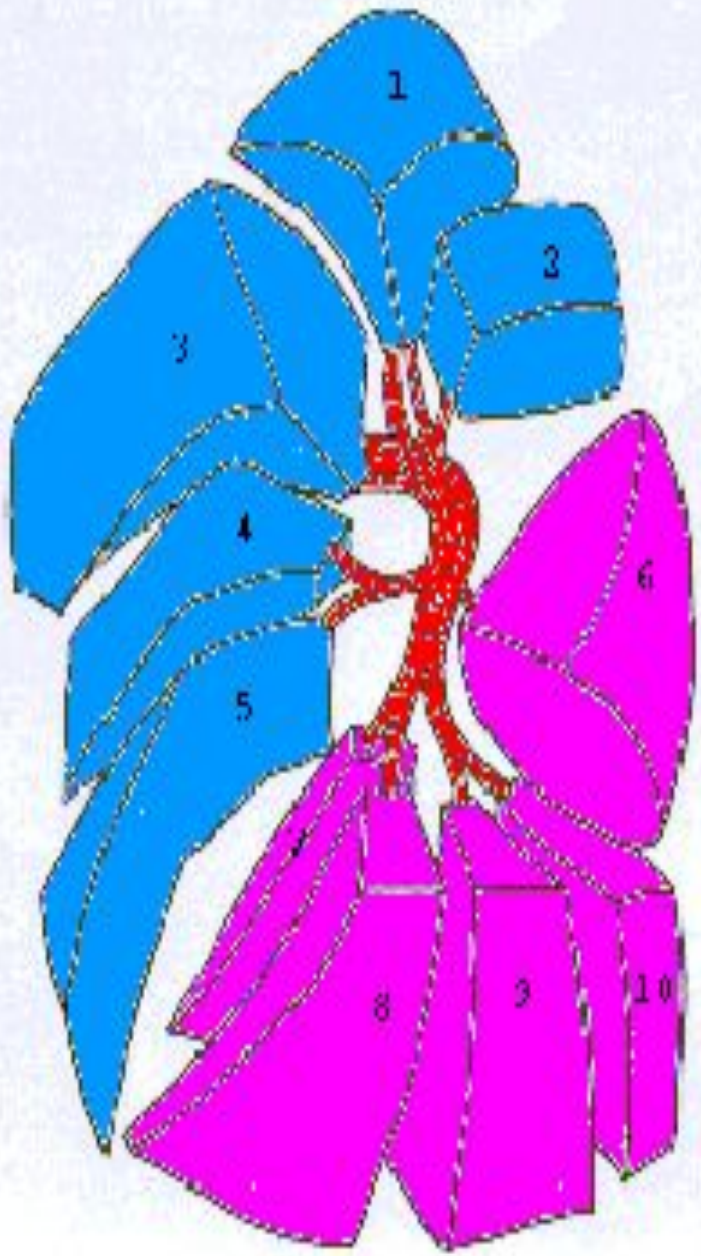
PULMONARY SEG AND BRONCHI



Right lung

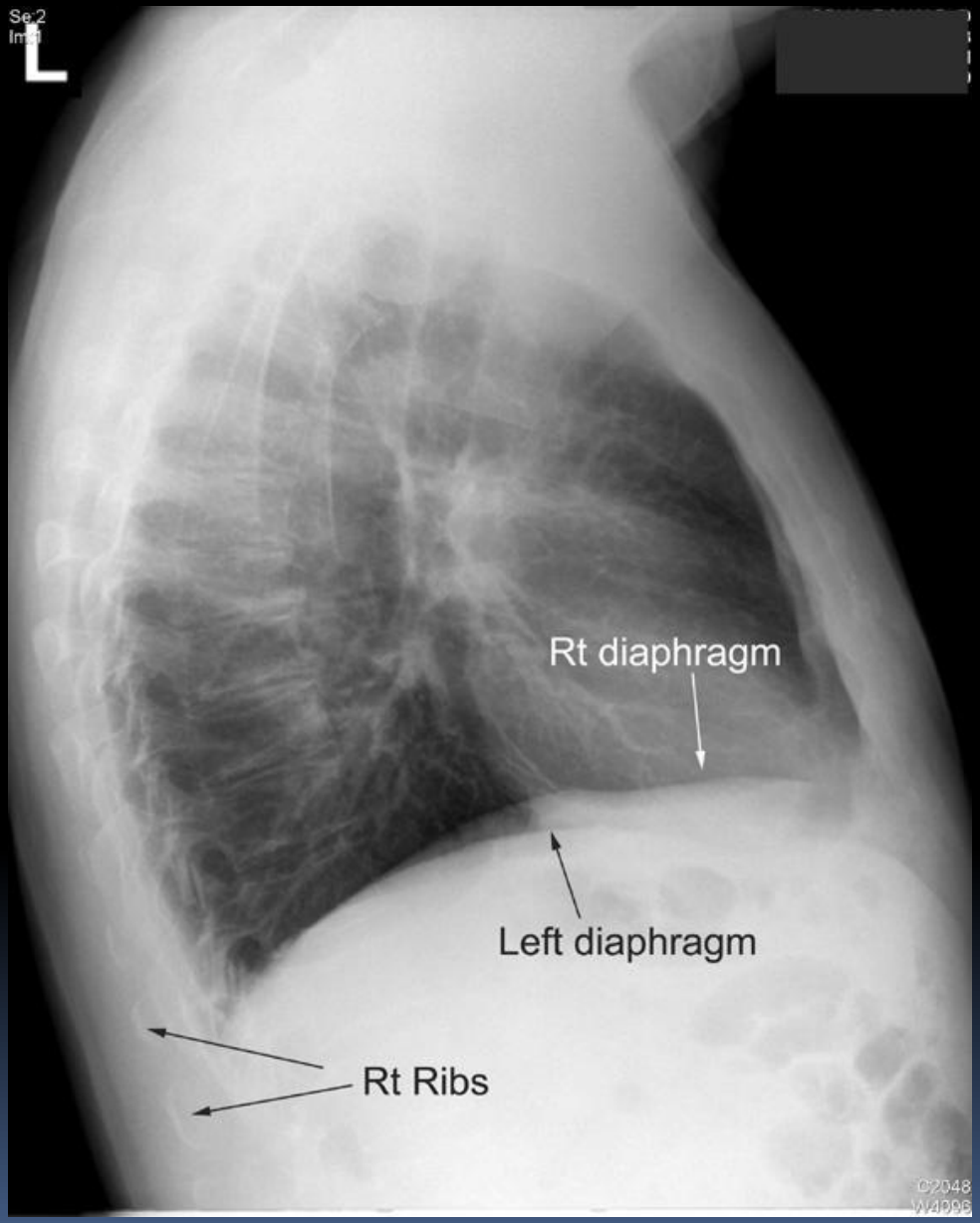


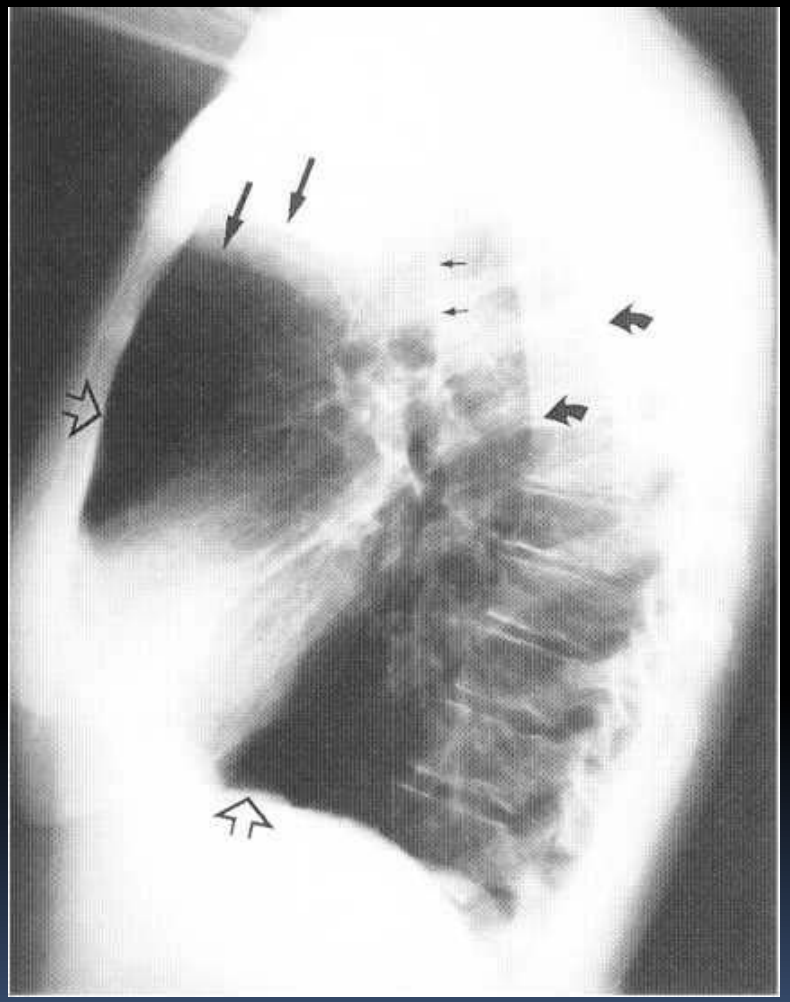
Left lung



lateral film

- Clear spaces-retrosternal and retrocardiac
- Vertebrae become more translucent caudally.
- Both diaphragms are visible throughout their length ,except left anteriorly.
- Axillary folds and scapulae overlie the lung fields.


Rt diaphragm
Left diaphragm
Rt Ribs

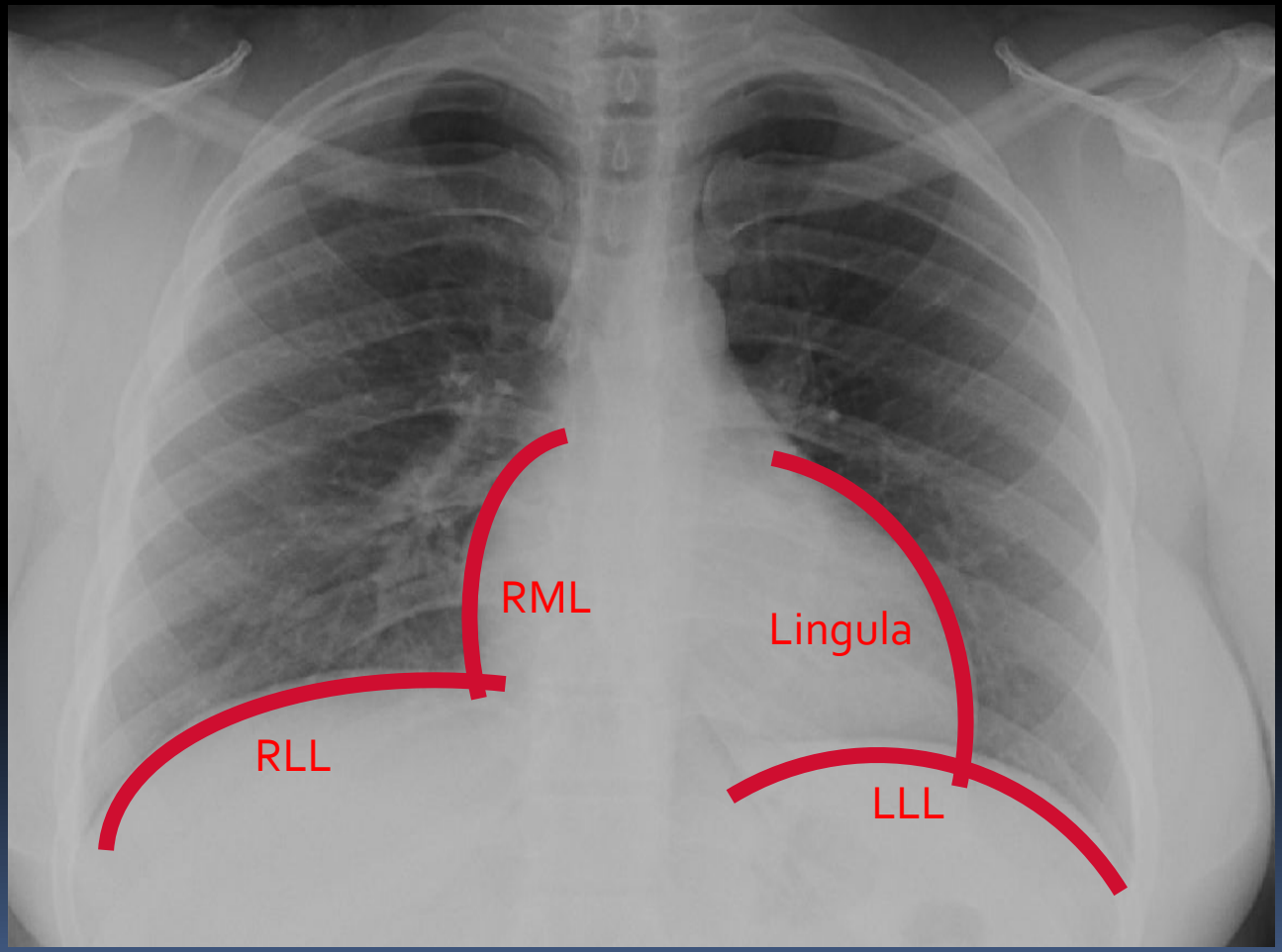


SILHOUETTE SIGN

AN INTRATHORASIC LESION TOUCHING A BORDER OF HEART,AORTA OR DIAPHRAGM WILL OBLITERATE THAT BORDER ON CXR.

AN INTRATHORASIC LESION NOT ANATOMICALLY CONTINUOUS WITH A BORDER OF ONE OF THESE STRUCTURES WILL NOT OBLITERATE THAT BORDER

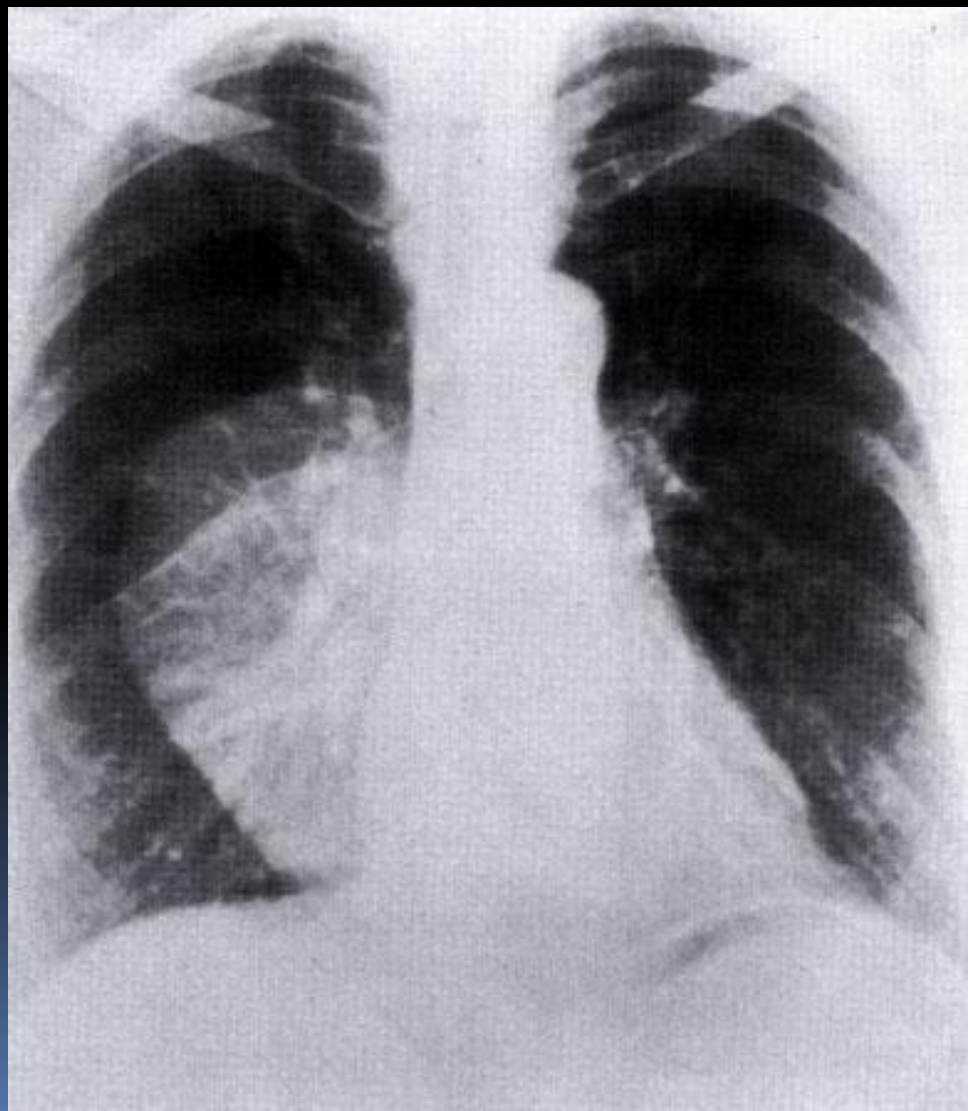
Localizing disease from the silhouette sign



Silhouette Sign

1) Radio-opacity that obliterate **part or all of the heart border**
   - Anterior Mediastinum

2) Radio-opacity that overlaps but does not obliterate **heart border**
   - Posterior Mediastinum



Silhouette Sign

3)Radio-opacity that obliterates **right border of ascending aorta**

- Anterior Mediastinum

4) Radio-opacity that overlaps but does not obliterate the **right borer of ascending aorta**

- Posterior Mediastinum



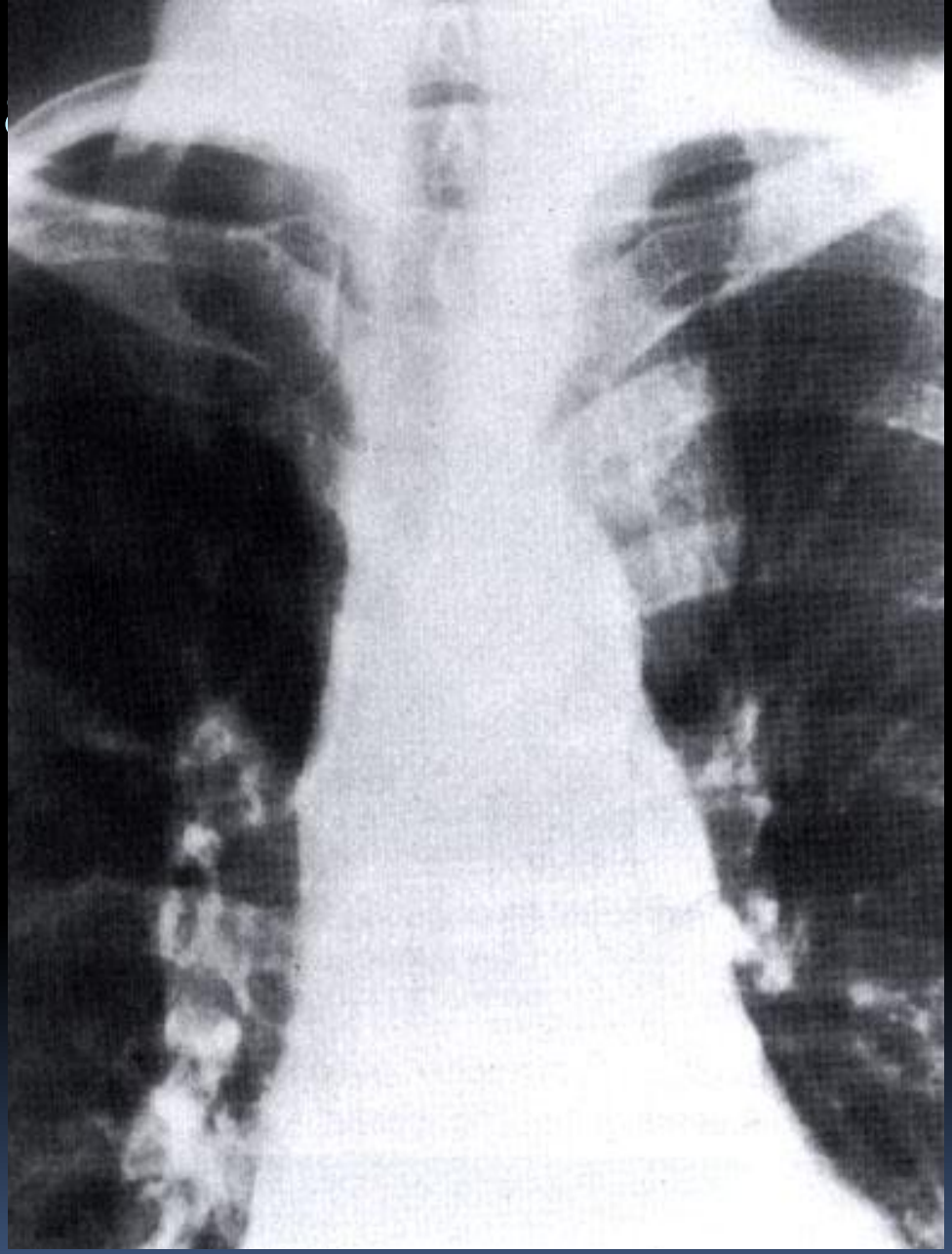
Silhouette Si

5) Radio-opacity that obliterates **left border of aortic knob**

- Posterior Mediastinum

6) Radio-opacity that overlaps but does not obliterate **aortic knob**

- Anterior Mediastinum or far Posterior Mediastinum



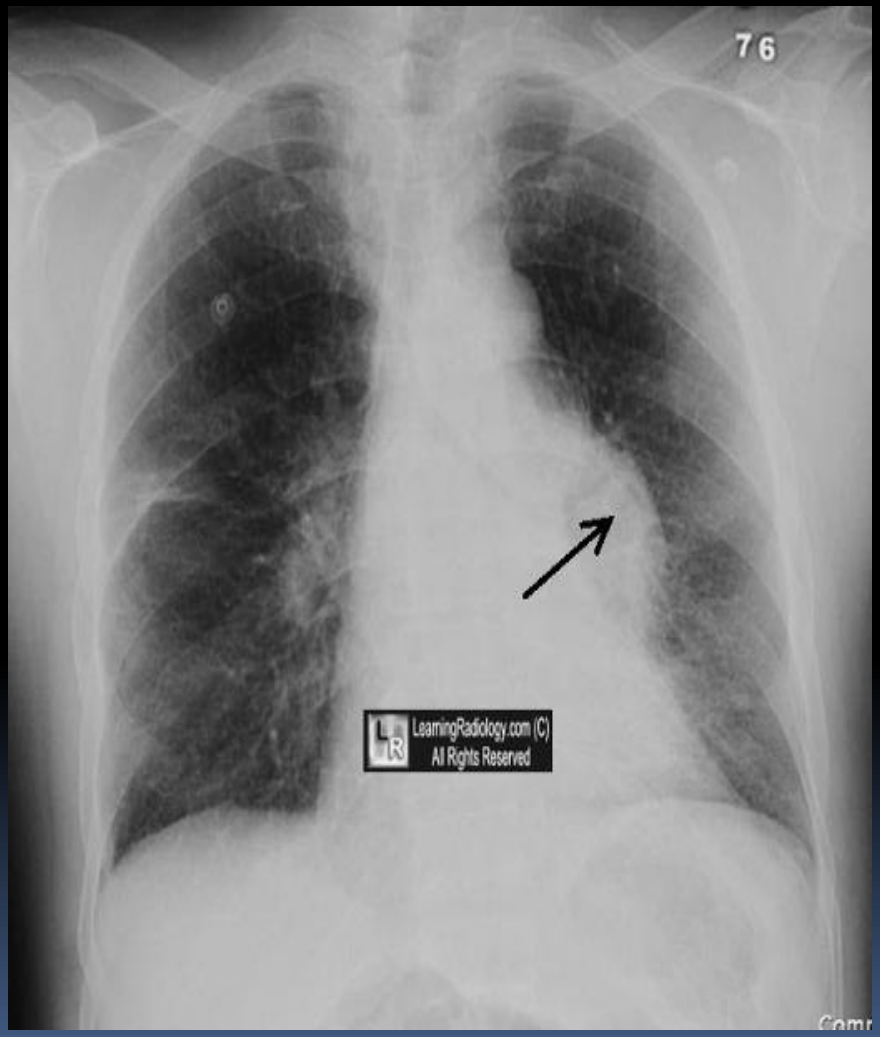
Hilum Overlay sign

- To differentiate between cardiomegaly & anterior mediastinal mass.



Hilum Convergence sign

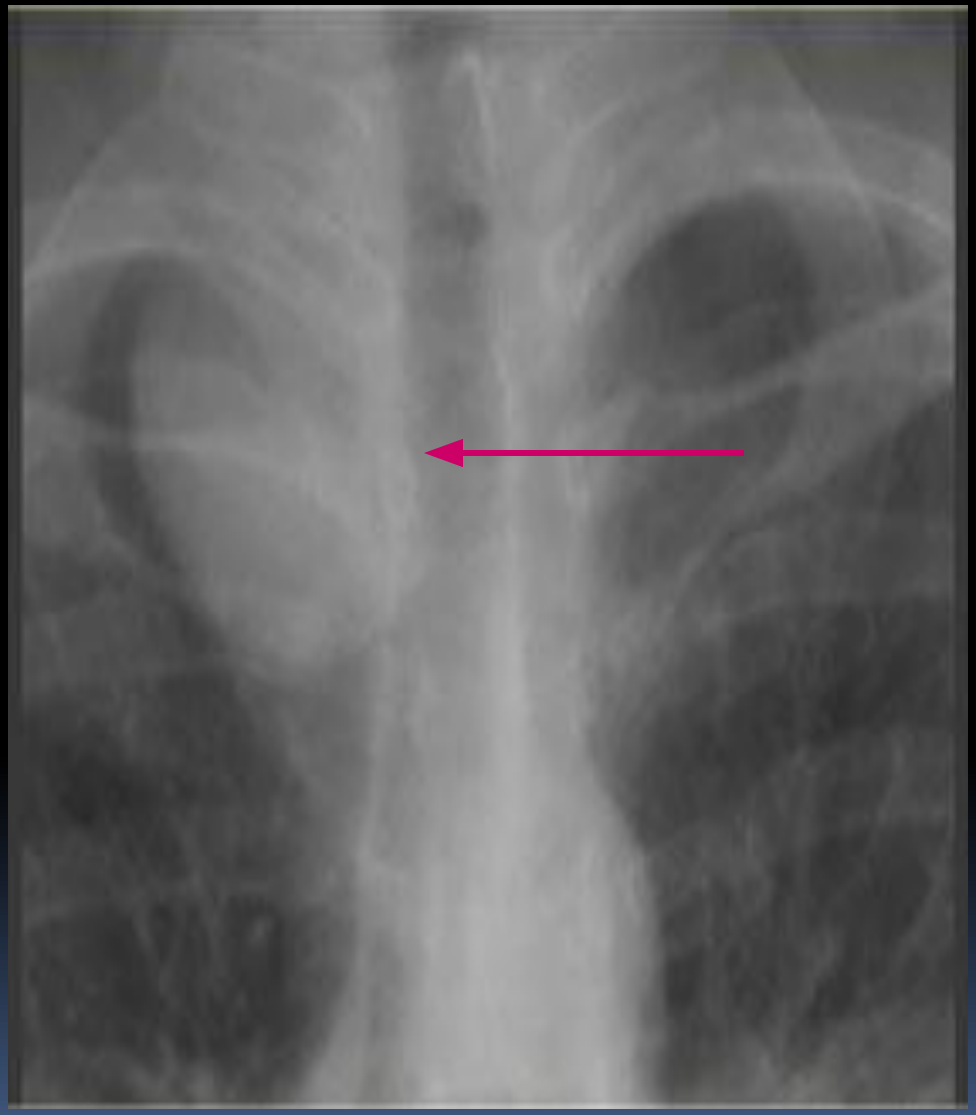
- To differentiate between enlarged pulmonary artery & juxta-hilar mass
- If the PA branches converge toward the mass rather than towards the heart it is an enlarged PA.



CERVICOTHORACIC SIGN

- Posterior mediastinal masses have well defined superior borders above the clavicle.
- Anterior masses are in contact with chest wall ,have ill defined margins.
- If the superior border merges with that of neck,it is ill defined and the lesion is both cervical and thoracic.

Cervico-thoracic sign



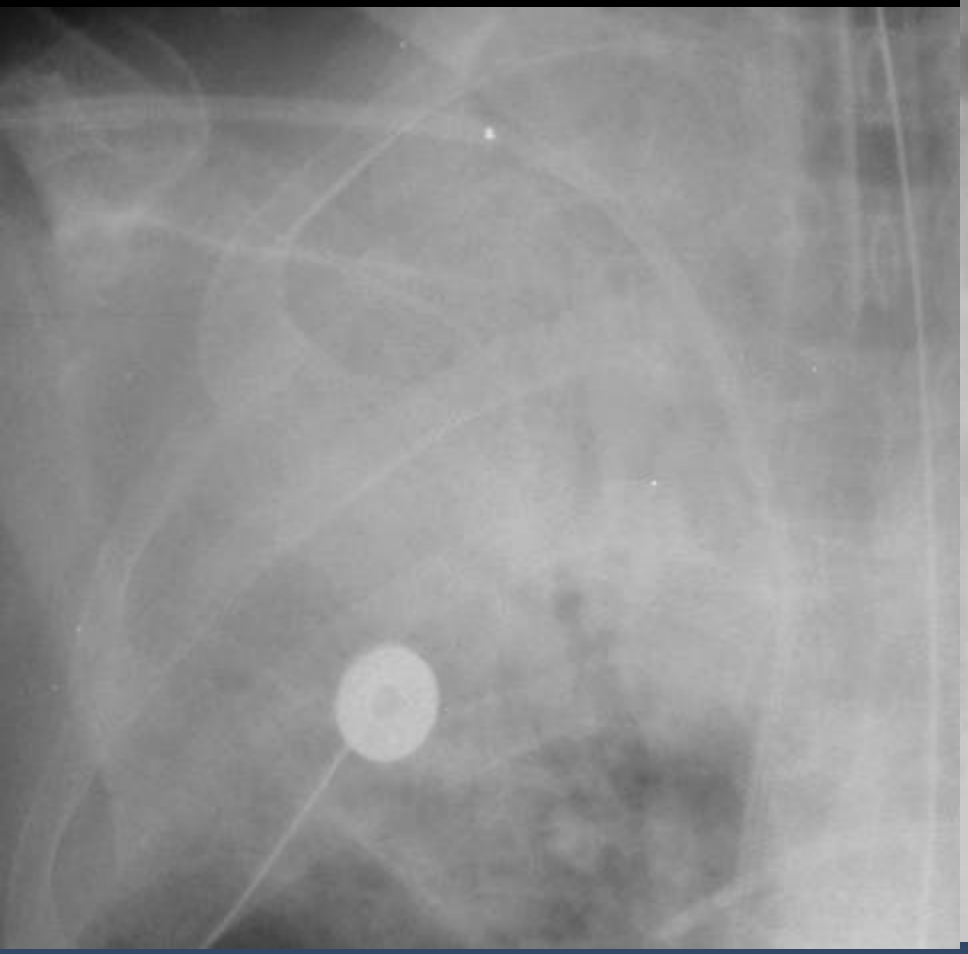
AIR BRONCHOGRAM SIGN

- Intrapulmonary bronchi are not visible on normal chest film,parenchymal consolidation may result in visualisation of these bronchi.

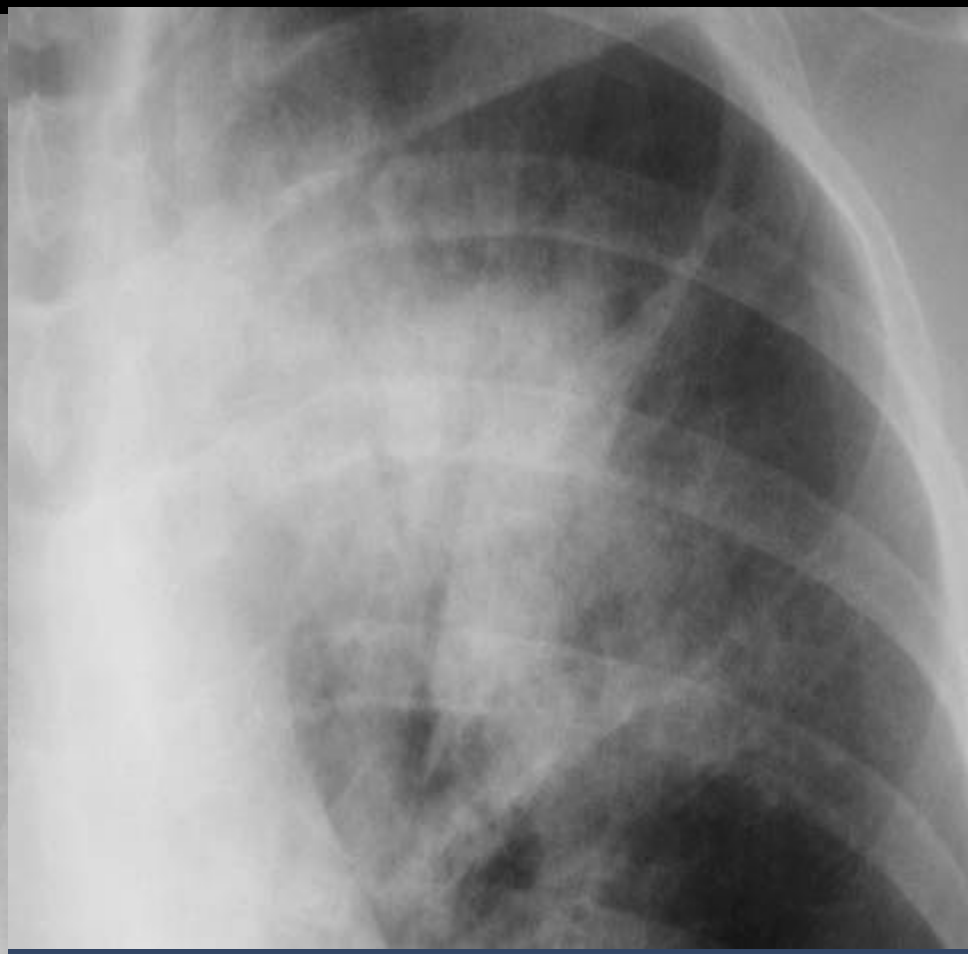
- SEEN AS SCATTERED LINEAR TRANSLUCENCIES.
- SEEN IN CONSOLIDATION ,MALIGNANCIES ETC.

- IMP SIGN SHOWING THE LESION IS INTRAPULMONARY

Air bronchogram sign



Pneumonia



Lung cancer

EXTRAPLEURAL SIGN

FEATURES:--

-PENSIL SHARP CONVEX OUTLINE

-TAPPERING MARGINS

SEEN IN:--

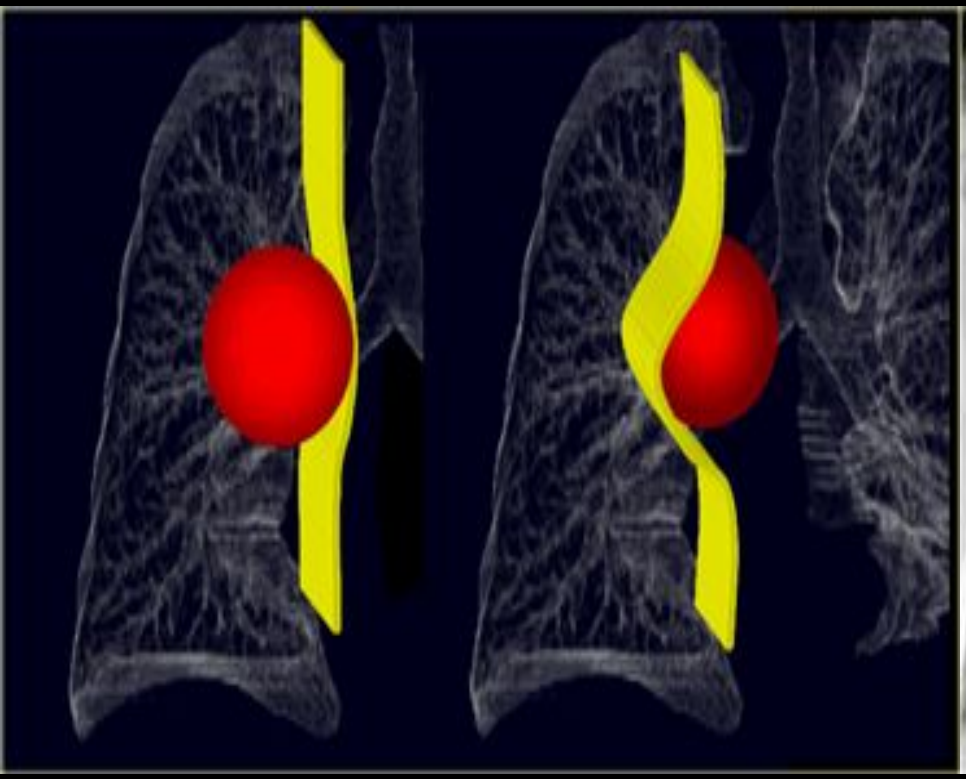
-LESIONS INVOLVING RIBS

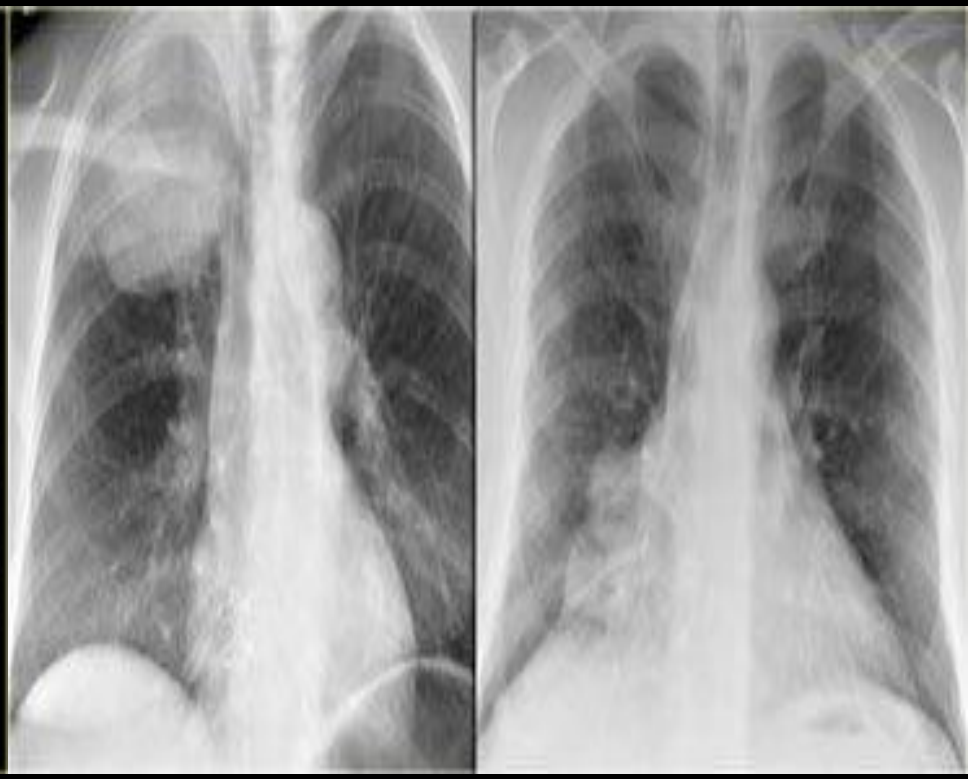
-MEDIASTINAL MASS

-CHEST WALL INFECTION



Intraparenchymal vs extraparenchymal





A lung mass abutts the mediastinal surface and creates acute angles with the lung, while a mediastinal mass will sit under the surface creating obtuse angles with the lung

ATELECTASIS or COLLAPSE

Signs of loss volume

Direct signs: 1) Opacity of the affected lung.

2) Crowding of vessels and bronchi within collapsed area.

3) Displacement or bowing of the fissures.

Indirect signs: 1) Compensatory hypertrophy of normal lungs or lobes resulting in an increased in transradiancy with separation of vascular markings.

2) Displacement of the mediastinal structures towards the affected side.

3) Displacement of ipsilateral hilum which changes shape

4) Elevation of ipsilateral hemi diaphragm.

5) Crowding of ribs on the affected side, particularly common in children.

RUL COLLAPSE

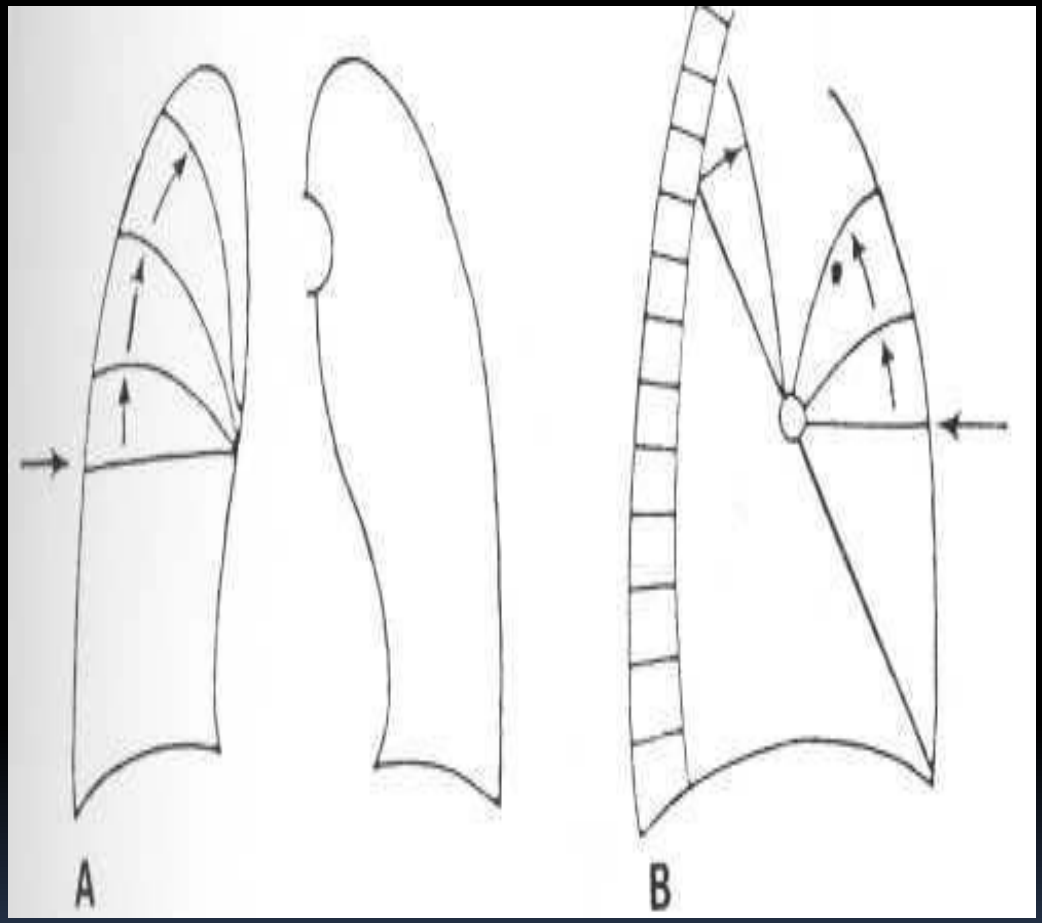
A) PA projection. lesser

fissure is drawn upward, and often curved, toward the apex and mediastinum.

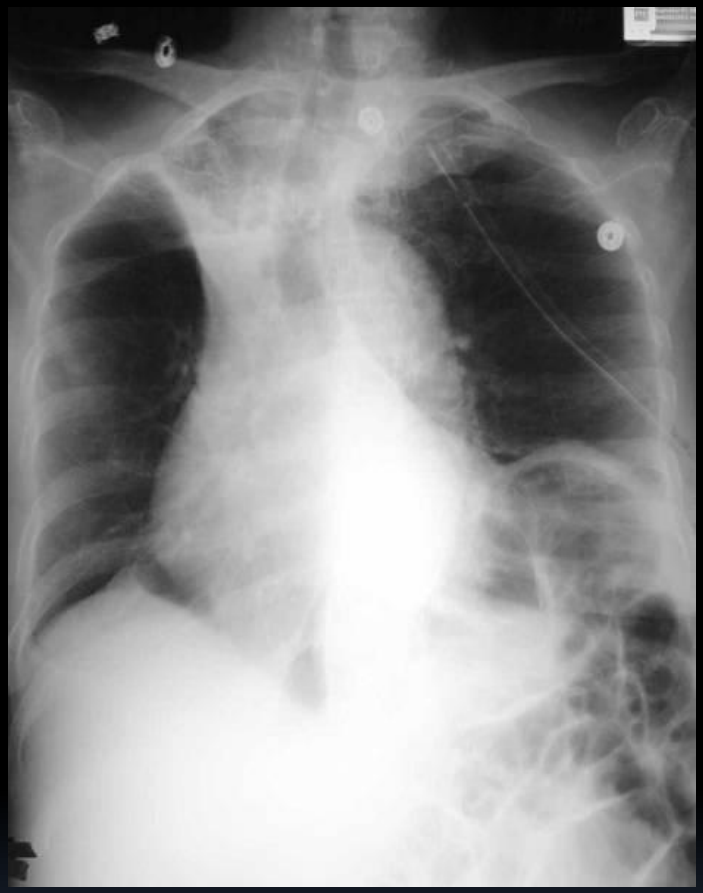
(B) Right lateral view. Lesser fissure also displaced upward. Note

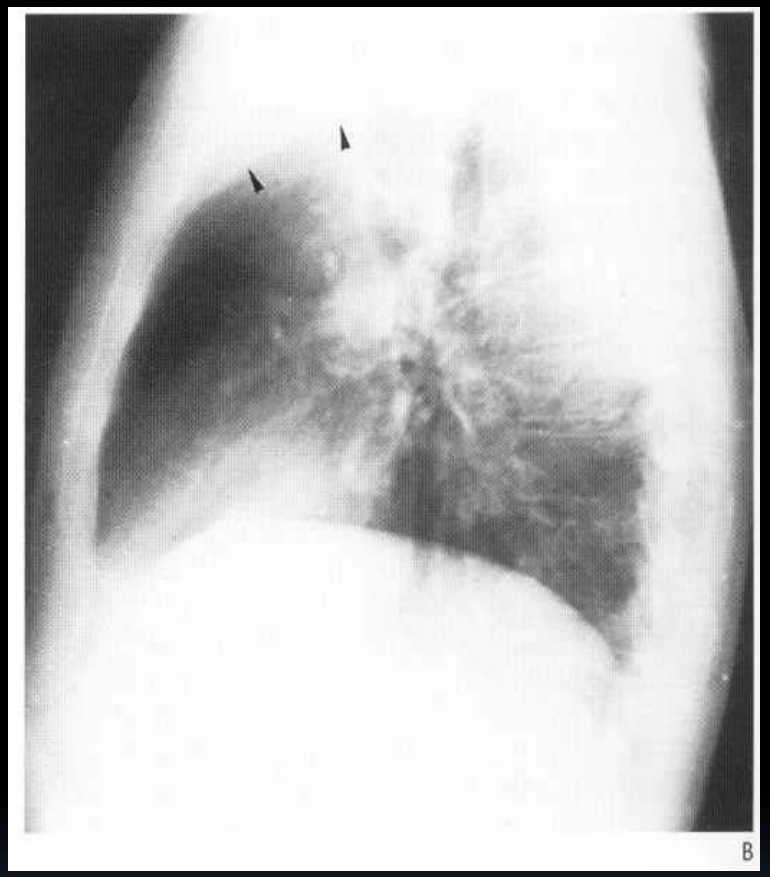
some forward displacement of greater fissure above the hilum

.



RUL ATELECTASIS

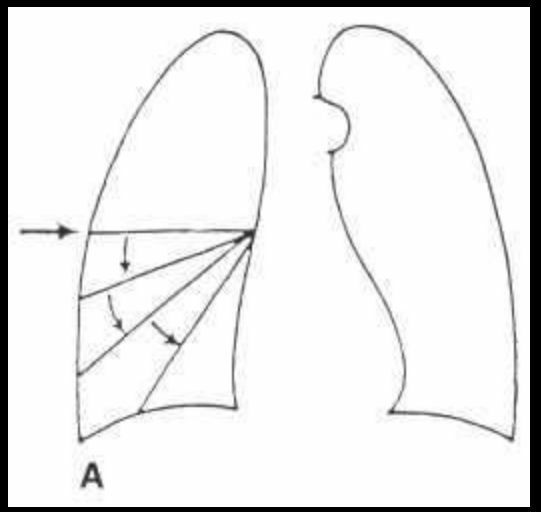


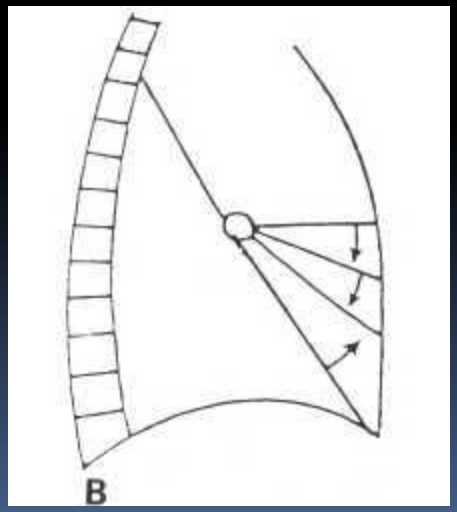


ELEVATION OF HORIZONTAL FISSURE, LAT-ELEVATION OF PART OF OBLIQUE FISSURE

RML COLLAPSE

Right middle lobe collapse
lobe collapse the horizontal fissure and lower half of the oblique
fissure move toward one another. This can best be seen in the
lateral projection. The horizontal fissure tends to be more mobile,
and therefore usually shows greater displacement. Signs of right
middle lobe collapse are often subtle on the frontal projection,
since the horizontal fissure may not be visible, and increased
opacity does not become apparent until collapse is almost complete.
However, obscuration of the right heart border is often



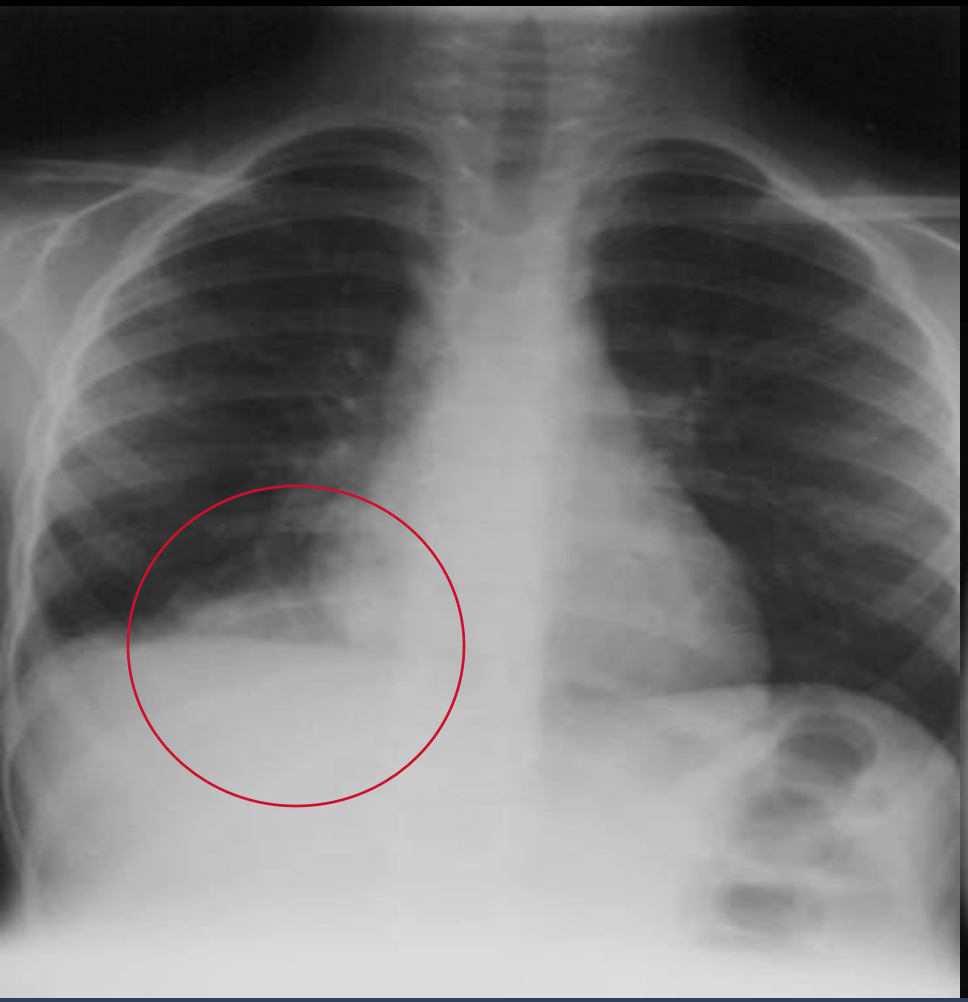


RML ATELECTASIS



LOSS OF DEFINATION OF RT HEART BORDER, LAT-WEDGE SHAPED OPACITY

RLL ATELECTASIS





Posterior diaphragm silhouetted

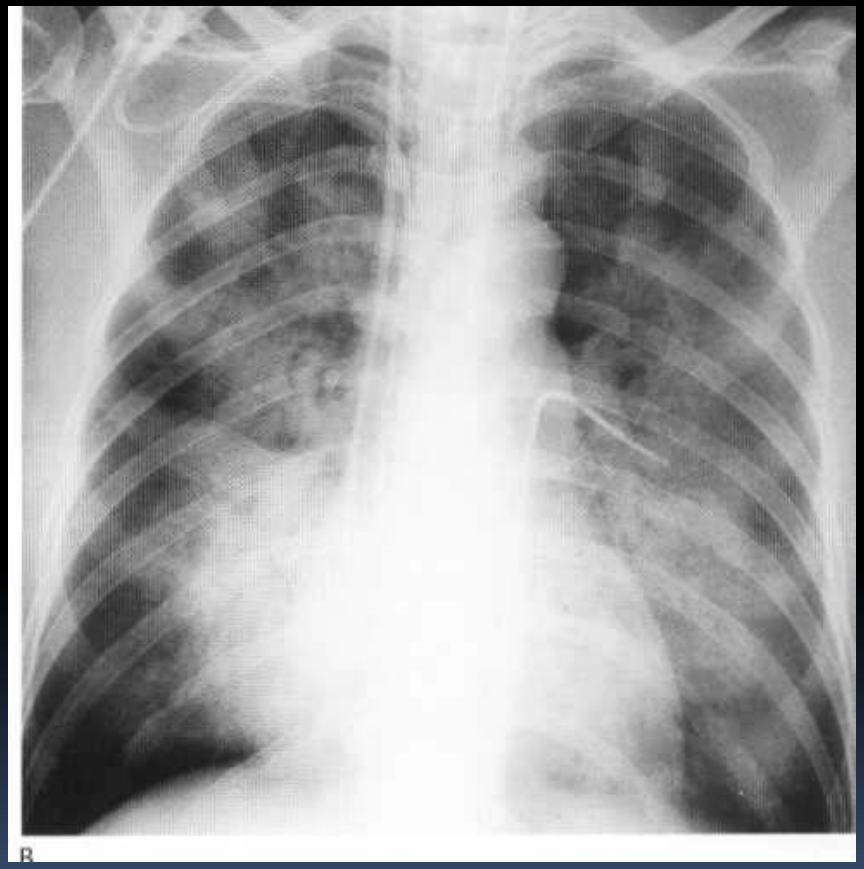
ELEVATION OF RT HEMIDIPHRAGM,DEPRESSION OF RT HILUM

Consolidation

Replacement of air by fluid in one or more acini.

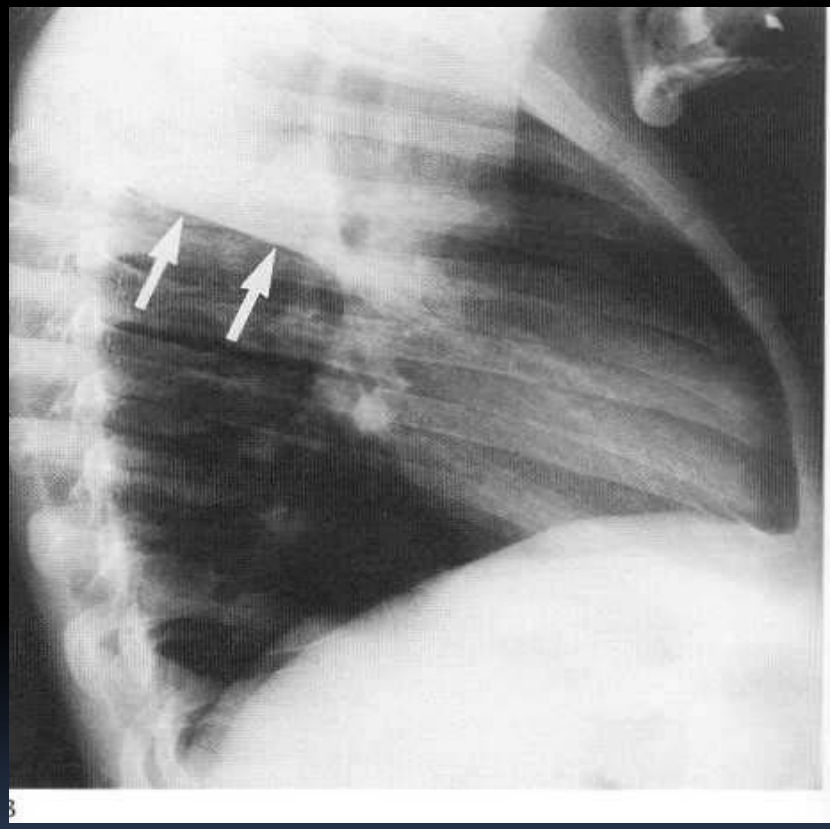
- Most common cause is inflammatory exudates in pnumonia
- Other causes are pulmonary edema ,neoplasm, aspiration etc.

consolidation



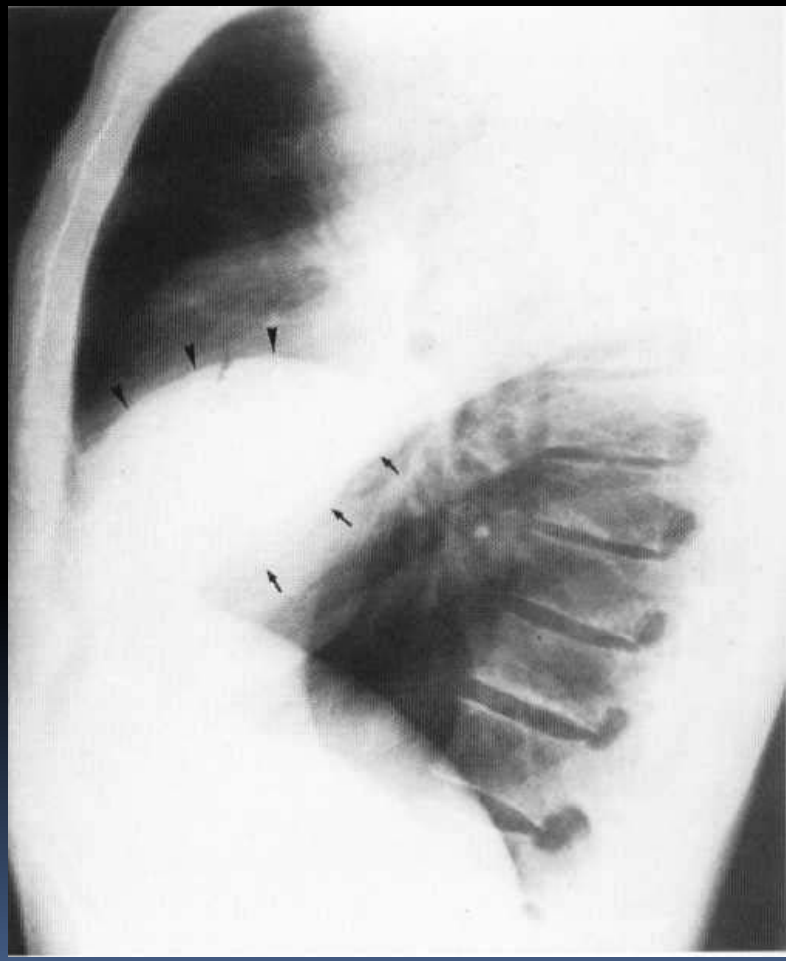
UPPER ZONE CONSOLIDATION





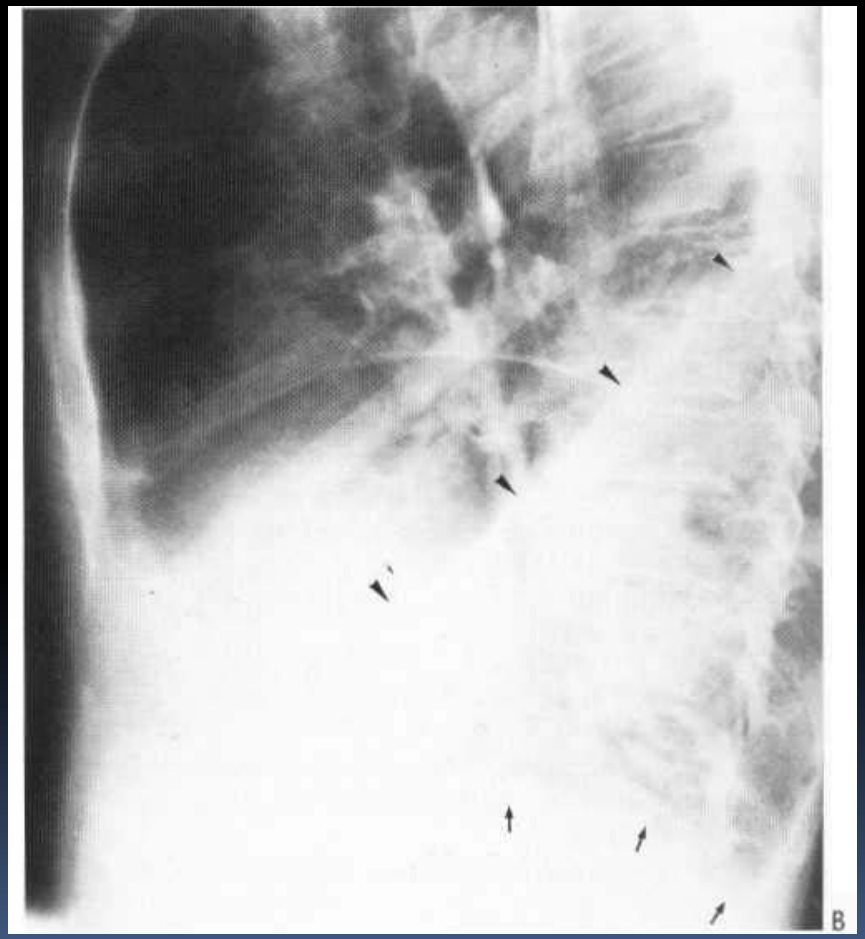
MID ZONE CONSOLIDATION





LOWER ZONE CONSOLIDATION





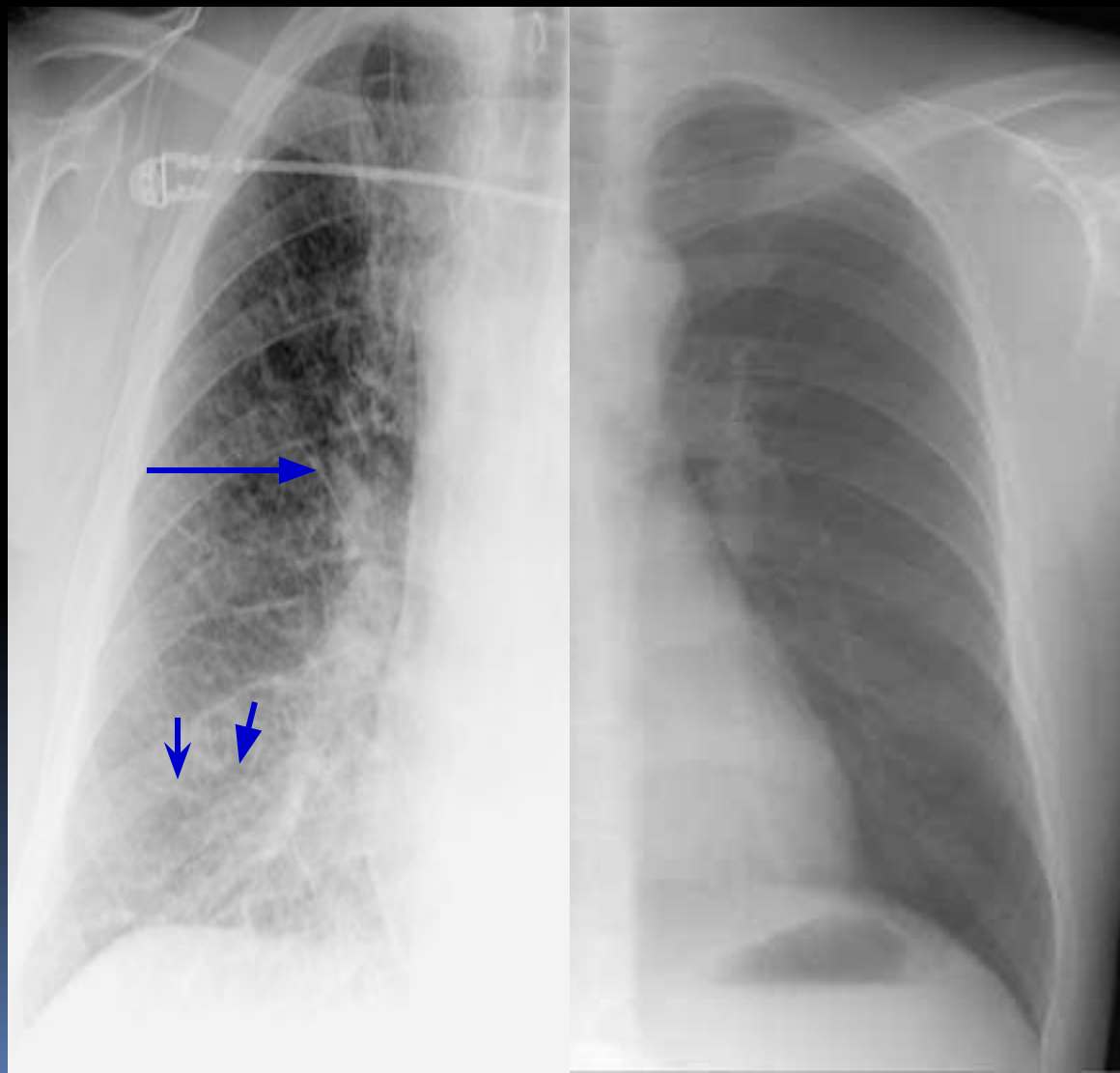
PULMONARY EDEMA.

SEPTAL LINES (KERLEY) .

KERLEY A LINES:

1-3MM WIDE 20-30MM LONG ;
NON BRANCHING,radiate from hila
THICKENING OF DEEP SEPTA

KERLEY B LINES:
1-2MM AND 20MM LONG;
NON BRANCHING, SHORT ,STRAIGHT PERIPHERAL LINES,perpendicular to pleura
THICKENING OF INTERLOBULAR SEPTA



PLEURAL EFFUSION

Small effusion collect in post cp angle,100-200 ml needed.

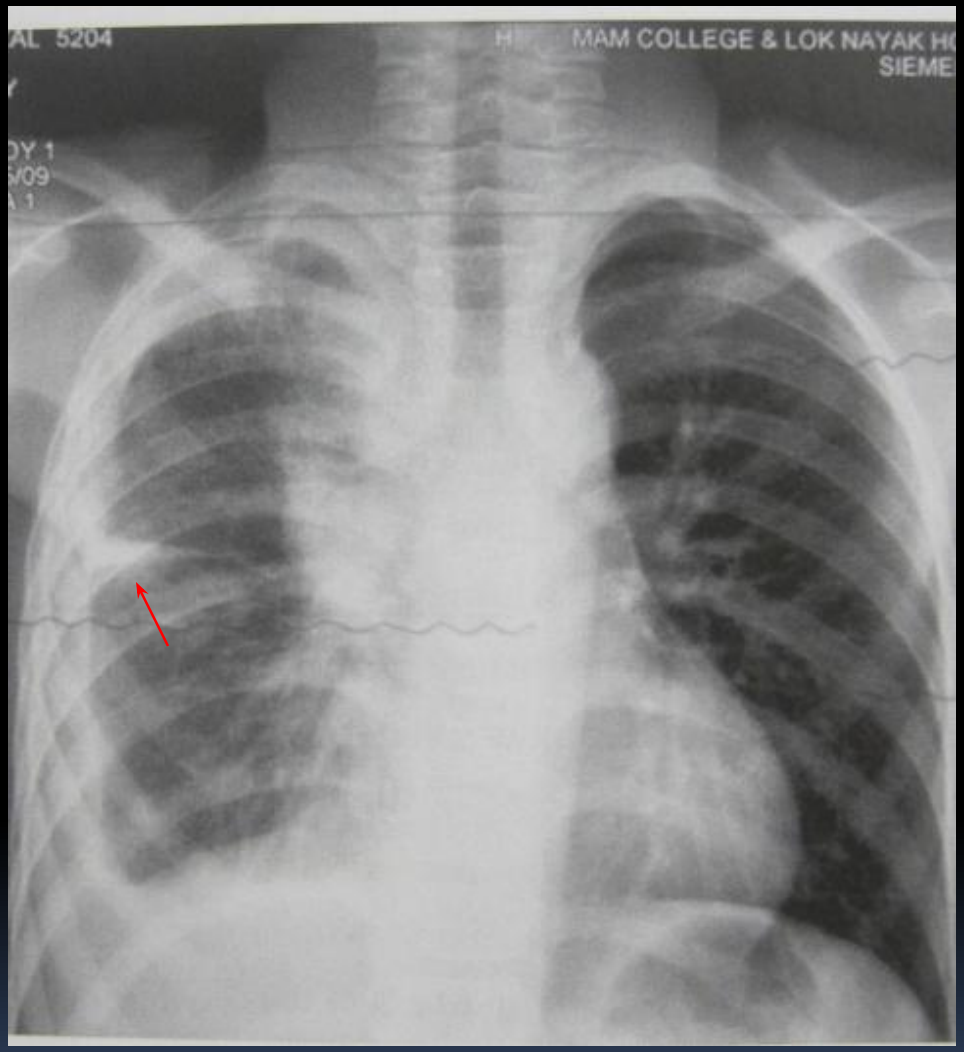
Lat decubitus -10 ml

signs

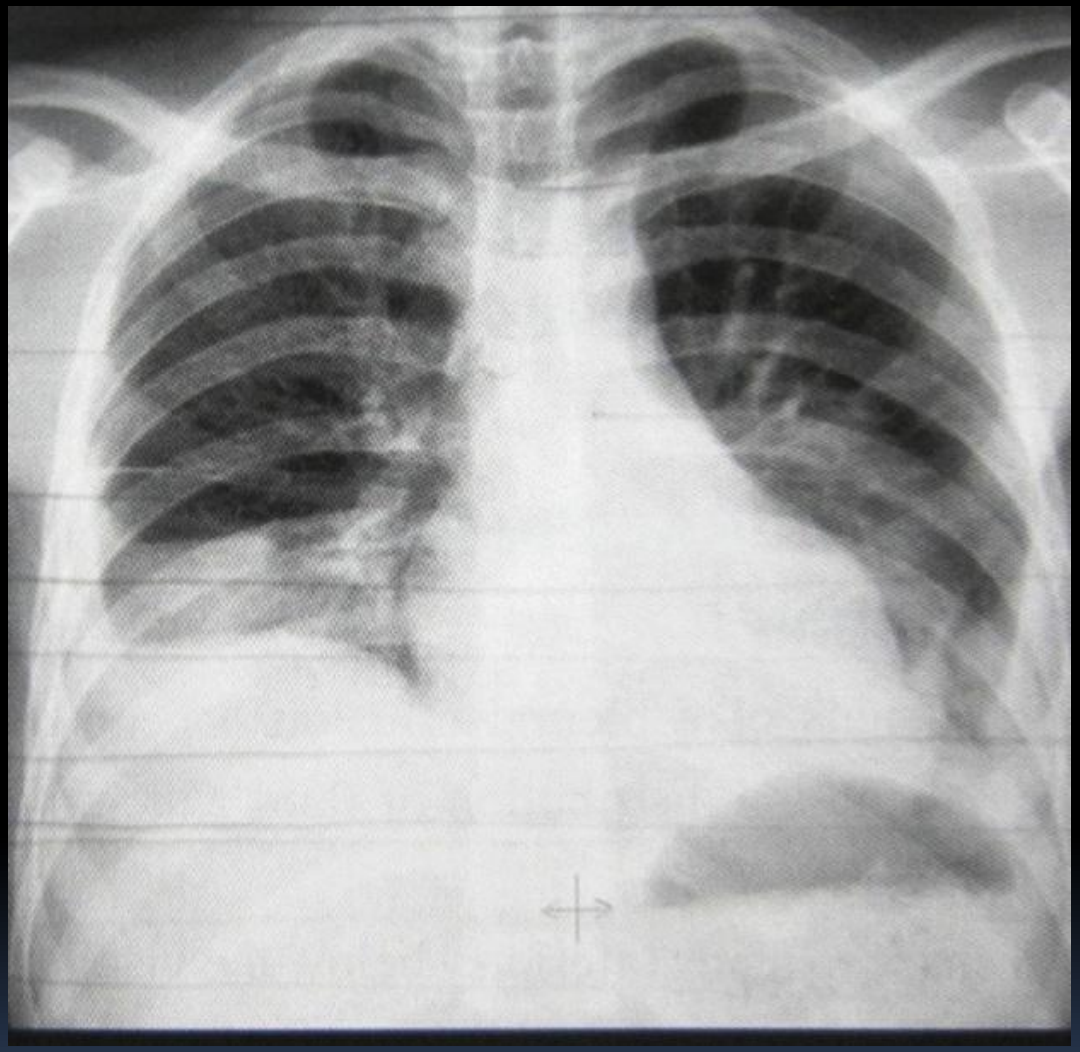
- Meniscus sign-Homogenous opacification of lower zone , obliteration of cp angle ,upper margin concave and higher laterally
- Thorn sign-fluid entering the minor fissure
- Raised rt heidiaphragm with lateralised apex
- Lamellar-vertical band of soft tissue density between lung and chest wall above cp angle



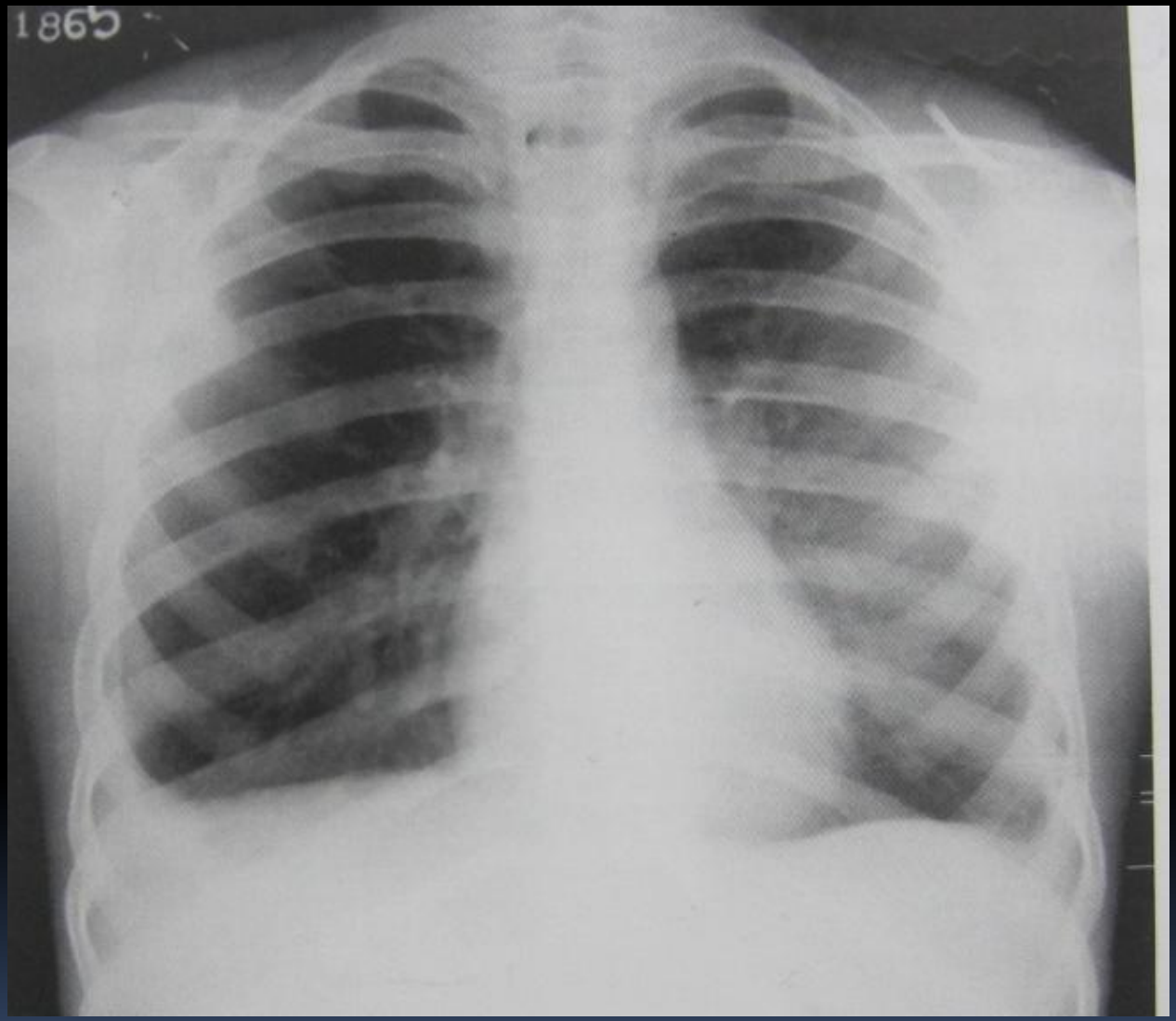
MENISCUS SIGN



THORN SIGN



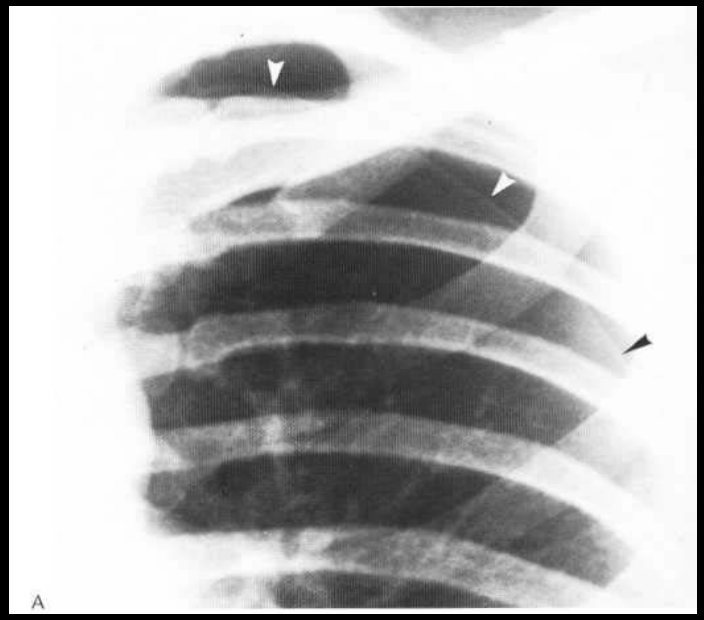
LATERALISED APEX



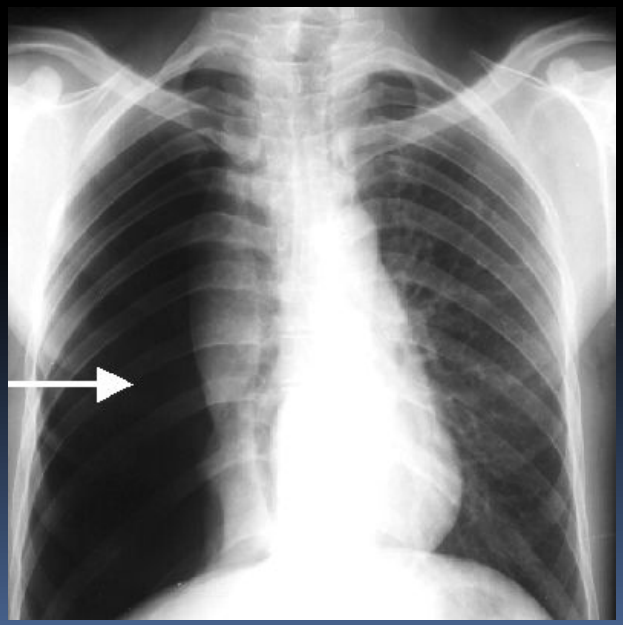
LAMELLAR EFFUSION

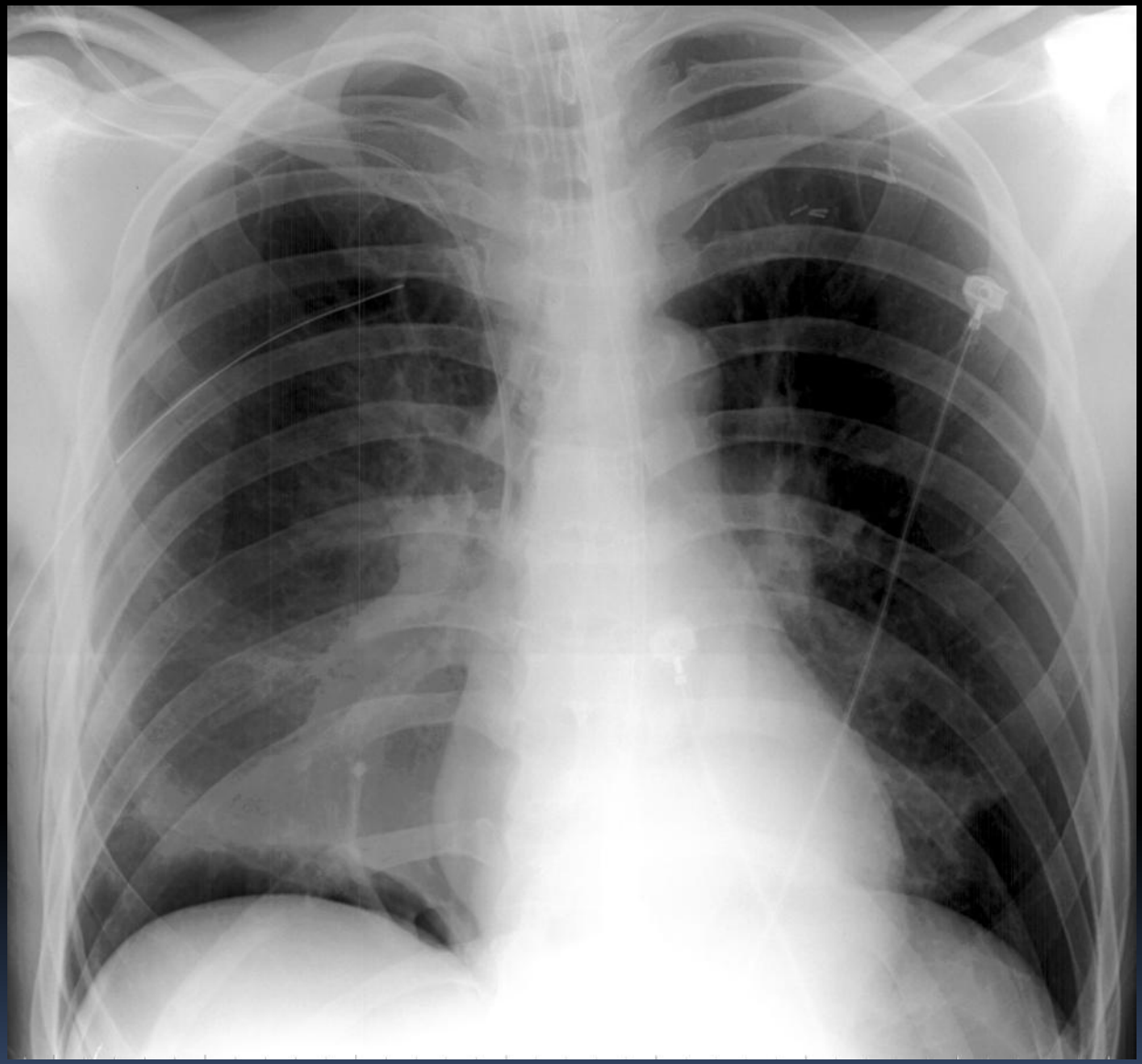
PNEUMOTHORAX

- Supine pt - ant pleural space
- Deep sulcus sign-cp angle deep and lucent
- Increased lucency over chest &abdomen
- Double diaphragm sign - Ant cp angle visualised as edge separate from diaphragm but parallel to it
- Increased sharpness


A







DOUBLE DIAPHRAGM SIGN

THANK YOU

